

IDENTIFICATION

PRODUCT CODE:	MAINDEC-08-DHKMC-B-D
PRODUCT NAME:	PDP-8E EXTENDED MEMORY ADDRESS TEST (EASE)
DATE RELEASED:	MAY 1976
MAINTAINER:	DIAGNOSTIC GROUP
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1. ABSTRACT

MODIFIED TO RUN ON APT 8A SYSTEMS, APRIL 1975.
SEE NOTES AT END OF DOCUMENT.

MODIFIED TO RUN WITH A NEW CONSOLE PACKAGE MAY 1976.
SEE SECTION 10 FOR CONSOLE ADDENDUM.

MODIFIED TO RUN WITH NO CONSOLE TERMINAL MAY 1976.
SEE SECTION 11 FOR OPERATING PROCEDURES ON A NON CONSOLE SYSTEM.

THE PDP-8E EXTENDED MEMORY ADDRESS TEST IS DESIGNED TO DETECT ANY LOCATION THAT CANNOT BE UNIQUELY ADDRESSED. THIS IS PERFORMED BY A SERIES OF FOUR TEST ROUTINES WHICH WILL TEST SYSTEMS EQUIPPED WITH FROM 8K TO 32K WORDS OF CORE MEMORY. AUTOMATIC PROGRAM RELOCATION IS PROVIDED IN ORDER TO TEST ALL MEMORY FIELDS FROM EACH MEMORY FIELD. TELETYPE PRINT-OUTS ARE PROVIDED FOR ERROR IDENTIFICATION, AND THE OPERATOR IS GIVEN A DEGREE OF CONTROL OVER THE PROGRAM BY VARIOUS SR SETTINGS.

2. REQUIREMENTS

2.1 EQUIPMENT

A PDP-8E OR A PDP-8A COMPUTER EQUIPPED WITH A MINIMUM OF 8K WORDS OF CORE MEMORY.

2.2 STORAGE

THE PROGRAM OCCUPIES CORE LOCATIONS 0000 TO 4777, WITH LOCATIONS 5000 TO 5177 USED AS A BUFFER AREA.

2.3 PRELIMINARY PROGRAMS

THE BINARY LOADER MUST BE IN MEMORY. ALSO, ALL DIAGNOSTICS FOR A BASIC 4K PDP-8E OR PDP-8A MUST HAVE BEEN PREVIOUSLY RUN SUCCESSFULLY.

3. LOADING PROCEDURE

LOAD THE PROGRAM WITH THE BINARY LOADER (BIN). THE PROGRAM MAY BE LOADED INTO ANY DESIRED CORE STACK BY HAVING BIN IN THAT CORE STACK.

4. OPERATING PROCEDURE

4.1 PROGRAM AND OPERATOR ACTION

- A. SET THE SR TO THE INSTRUCTION FIELD AND DATA FIELD OF THE STACK WHICH CONTAINS THE PROGRAM.
- B. PRESS KEY EXT0 ADDR LOAD.
- C. SET THE SR FOR DESIRED STARTING ADDRESS ACCORDING TO THE FOLLOWING TABLE.

ADDRESS	TEST EXECUTION
0200	RUN ALL TESTS
0201	RUN ONLY TEST 1
0202	RUN ONLY TEST 2
0203	RUN ONLY TEST 3
0204	RUN ONLY TEST 4

- D. PRESS KEYS ADDR LOAD, CLEAR, AND CONT. A SETUP SR MESSAGE WILL BE PRINTED.

- E. SET THE SR FOR DESIRED OPERATION ACCORDING TO THE FOLLOWING TABLE.

SWITCH	0 (DOWN)	1 (UP)
SR00	CONTINUE AFTER ERROR	HALT AFTER ERROR
SR01	TYPEOUT ERRORS	INHIBIT ERROR TYPEOUTS
SR02	NORMAL	TTY BELL ON ERROR
SR03	RELOCATE PROGRAM	INHIBIT PROGRAM RELOCATION
SR04	NORMAL	CHANGE STACK LIMITS
SR05	NORMAL	HALT AFTER CURRENT TEST
SR06-08	STARTING STACK LIMIT (0-7)	
SR09-11	ENDING STACK LIMIT (0-7)	

- F. PRESS KEY CONT.

4.2

DETAILED SR EXPLANATION

SR00-02 SR02, WILL RING THE TTY BELL ONCE FOR EACH ERROR.
SR00 AND SR01 HAVE NO EFFECT WITH SR02 SET.
SR03 SR03 MAY BE SET OR RESET AT ANY TIME AND THE PROGRAM
WILL ACT ACCORDINGLY
SR04 SR04 ALLOWS THE OPERATOR TO CHANGE THE STACK LIMITS AS
DEFINED BY SR06-11.
SR05 SR05 IS NORMAL HALT FOR PROGRAM
SR06-08 THESE SWITCHES DEFINE THE STARTING STACK LIMIT
(NORMALLY 0).
SR09-11 THESE SWITCHES DEFINE THE ENDING STACK LIMIT
(NORMALLY 7)

4.3

EXAMPLE OF SELECTING STACKS FOR TEST

EXAMPLE 1: SR = 0007, 28K SYSTEM

STACKS SELECTED FOR TESTING ARE 6,5,4,3,2,1,0

EXAMPLE 2: SR = 0004, 28K SYSTEM

STACKS SELECTED FOR TESTING ARE 4,3,2,1,0

EXAMPLE 3: SR = 0022 28K SYSTEM

STACKS SELECTED FOR TESTING ARE 2
(NO RELOCATION WILL OCCUR)

EXAMPLE 4: SR = 0041 28K SYSTEM

STACKS SELECTED FOR TESTING ARE 6,5,4,1,0

NOTE 1: STACKS NOT IN THE SYSTEM ARE AUTOMATICALLY DE-SELECTED
AS IS EXAMPLE 1. STACK 7 IS NOT PRESENT THEREFORE NOT
SELECTED.

NOTE 2: A SINGLE STACK CAN BE SELECTED FOR TESTING PROVIDING
THE PROGRAM IS NOT IN THAT STACK AS IN EXAMPLE 3.

NOTE 3: ANY STACK OR GROUP OF STACKS CAN BE BY-PASSED AS IN
EXAMPLE 4. STACKS 2 AND 3 ARE NOT SELECTED, STACK 7
IS NOT PRESENT.

5.

ERRORS

THE CONTENTS OF A GIVEN MEMORY TEST LOCATION SHOULD ALWAYS BE
EQUAL TO ITS ADDRESS OR THE COMPLEMENT OF ITS ADDRESS. IF IT
IS NOT, A TEST ERROR WILL RESULT. A RELOCATION ERROR WILL
OCCUR IF THE RELOCATION COMPARISON CHECK FAILS.

5.1

TEST ERROR TYPEOUTS -----

FOR THE FIRST ERROR ENCOUNTERED A HEADER WILL BE TYPED OUT FOLLOWED BY THE PERTINENT DATA. FOR ALL SUBSEQUENT ERRORS, ONLY THE PERTINENT DATA WILL BE TYPED. THE FORMAT IS AS FOLLOWS:

PR LOC ADDR GOOD BAD TEST

PR LOC = THE PROGRAM ADDRESS WHERE THE ERROR JMS OCCURRED.
(INCLUDES FIELD)

ADDR = THE ADDRESS OF THE LOCATION IN ERROR. (INCLUDES FIELD)

GOOD = WHAT THE DATA SHOULD BE.

BAD = WHAT THE DATA IS.

TEST = THE TEST (1-4) RUNNING WHEN THE FAILURE OCCURRED.

5.2

RELOCATION ERROR TYPEOUTS -----

ALL RELOCATION ERRORS ARE IN THE FOLLOWING FORMAT:

XXXXX RELOCATION ERROR AT LOCATION YYYY

XXXXX = THE PROGRAM ADDRESS WHERE THE ERROR JMS OCCURRED.
(INCLUDES FIELD)

YYYYY = THE ADDRESS OF THE LOCATION IN ERROR (INCLUDES FIELD)

NOTE: AFTER EACH ERROR PRINT-OUT THE PROGRAM CONTINUES
ON WITH THE NEXT SEQUENTIAL MEMORY LOCATION.

6.

RESTRICTIONS -----

6.1

STARTING RESTRICTIONS -----

THE PROGRAM MAY BE RESTARTED AT ANY TIME FROM LOCATION 0200
OF THE STACK THE PROGRAM IS PRESENTLY IN.

6.2 OPERATING RESTRICTIONS

NONE

7. EXECUTION TIME

THE TIME TO RUN ALL 4 TESTS IN ONE CORE STACK IS APPROXIMATELY 1/2 SECOND. DURING PROGRAM EXECUTION A 5 WILL BE TYPED ON THE TTY APPROXIMATELY EVERY 5 MINUTES OF PROGRAM RUN TIME. THIS ALLOWS THE OPERATOR TO DETERMINE APPROXIMATE RUN TIME BEFORE A FAILURE OCCURRED.

8. SCOPE LOOPS

TWO SPECIAL SCOPE LOOPS HAVE BEEN PROVIDED IN THIS PROGRAM.

8.1 SCOPE LOOP 1

THIS SCOPE LOOP WRITES THE VALUE EQUAL TO THE ADDRESS SPECIFIED BY THE SR INTO THE ADDRESS SPECIFIED BY THE SR. IT THEN LOOPS DOING A WRITE-READ. THE ADDRESS BEING LOOPED ON CAN BE CHANGED SIMPLY BY CHANGING THE SWITCH SETTING.

- A. SET THE SR TO THE INSTRUCTION FIELD THAT THE PROGRAM IS IN AND THE DATA FIELD WANTED TO TEST.
- B. PRESS KEY EXT0 ADDR LOAD.
- C. SET THE SR EQUAL TO 3200.
- D. PRESS KEY ADDR LOAD.
- E. SET THE SR EQUAL TO THE ADDRESS TO TEST.
- F. PRESS KEYS CLEAR, AND CONT.

8.2 SCOPE LOOP 2

THIS SCOPE LOOP IS THE SAME AS SCOPE LOOP 1 EXCEPT THAT A GROUP OF ADDRESSES MAY BE SPECIFIED. THE STARTING ADDRESS SPECIFIED MUST BE LESS THAN THE ENDING ADDRESS SPECIFIED.

- A. SET THE SR TO THE INSTRUCTION FIELD THAT THE PROGRAM IS IN AND THE DATA FIELD WANTED TO TEST.
- B. PRESS KEY EXT0 ADDR LOAD.
- C. SET THE SR EQUAL TO 3207.

- D. PRESS KEY ADDR LOAD.
- E. SET THE SR EQUAL TO THE FIRST ADDRESS OF THE GROUP.

- F. PRESS KEYS CLEAR AND CONT. A HALT WILL OCCUR AT ADDRESS 3211.
- G. SET THE SR EQUAL TO THE LAST ADDRESS OF THE GROUP.

- H. PRESS KEY CONT.

NOTE 1: THE ADDRESS(S) SPECIFIED WILL BE LOOPED UNTIL STOPPED BY THE OPERATOR WITH KEY HALT. NO ERROR CHECKING IS DONE. TO RESUME NORMAL OPERATION, RESTART PROGRAM AT ADDRESS 0200-0204 OF THE CURRENT INSTRUCTION FIELD.

9. PROGRAM DESCRIPTION -----

9.1 GENERAL -----

THE PDP-8E EXTENDED MEMORY ADDRESS TEST IS INTENDED FOR USE WITH A PDP-8E EQUIPPED WITH THE EXTENDED MEMORY OPTION. A TOTAL OF FOUR TESTS ARE EXECUTED BY THE PROGRAM. (SEE 9.2 THRU 9.5). EACH TEST WRITES A UNIQUE PATTERN INTO CORE MEMORY AND THE CHECKS FOR ERROR. THE PATTERNS WERE CHOSEN TO AID THE OPERATOR IN THE EVENT OF ADDRESSING ERRORS.

THE PROGRAM AUTOMATICALLY RELOCATES ITSELF TO EACH MEMORY FIELD UNDER TEST TO ENSURE THAT ALL FIELDS MAY BE CORRECTLY REFERENCED FROM ANY FIELD. FIELDS NOT PRESENT IN THE SYSTEM WILL AUTOMATICALLY BE DE-SELECTED FROM TESTING. (SEE 9.6)

CONTROL OF THE PROGRAM IS GIVEN TO THE OPERATOR BY MEANS OF THE SR. THE OPERATOR MAY HALT AFTER ERROR, INHIBIT ERROR PRINTOUTS, SUBSTITUTE TTY BELL FOR ERROR INDICATION, HALT AFTER TEST, CHANGE FIELD TEST LIMITS, SELECT ALL OR ANY ONE OF FOUR TESTS, INHIBIT PROGRAM RELOCATION, AND AT ANY TIME RESTART THE PROGRAM AT LOCATION 0200 THRU 0204.

9.2 TEST 1 -----

TEST 1 WRITES THE VALUE OF EACH LOCATION INTO ITSELF IN THE FORWARD DIRECTION. THEN EACH LOCATION IS READ AND CHECKED IN THE FORWARD DIRECTION.

TEST 2

TEST 2 WRITES THE COMPLEMENT VALUE OF EACH LOCATION INTO ITSELF IN THE FORWARD DIRECTION. THEN EACH LOCATION IS READ AND CHECKED IN THE FORWARD DIRECTION.

TEST 3

TEST 3 WRITES THE VALUE OF EACH LOCATION INTO ITSELF IN THE REVERSE DIRECTION. THEN EACH LOCATION IS READ AND CHECKED IN THE REVERSE DIRECTION.

TEST 4

TEST 4 WRITES THE COMPLEMENT VALUE OF EACH LOCATION INTO ITSELF IN THE REVERSE DIRECTION. THEN EACH LOCATION IS READ AND CHECKED IN THE REVERSE DIRECTION.

PROGRAM RELOCATION

PROGRAM RELOCATION IS GOVERNED BY THE STATUS OF SR BIT 3 OR BY THE FACT THAT ONLY ONE STACK IS SELECTED FOR TESTING. WITH SR BIT 3 DOWN (0 POSITION) PROGRAM RELOCATION OCCURS EACH TIME THE TEST PATTERN AND ITS COMPLEMENT HAVE BEEN COMPLETELY TESTED IN EACH SELECTED STACK. THE PROGRAM FIRST RELOCATES TO THE HIGHEST ORDER 4K STACK UNDER TEST. THE PROGRAM KEEPS RELOCATING TO THE NEXT LOWER STACK UNDER TEST UNTIL IT REACHES THE LOWEST ORDER STACK UNDER TEST. THE TESTING AND RELOCATION CYCLE IS THEN REPEATED. THE CONTENTS OF THE ENTIRE STACK ARE RELOCATED WHICH ENABLES ANY OTHER INFORMATION (RIM-BIN) TO BE CARRIED WITH THE PROGRAM.

THE PROGRAM PROVIDES A DEGREE OF PROTECTION FOR ITSELF BY REMEMBERING ALL STACKS WHERE ERRORS OCCUR. WHEN A FAULTY STACK IS NEXT IN SEQUENCE TO CONTAIN THE PROGRAM, THE PROGRAM WILL SKIP THE FAULTY STACK AND RELOCATE TO THE FIRST LOWER ORDER STACK WHICH IS ERROR FREE. IF ALL OTHER SELECTED STACKS ARE FAULTY, PROGRAM RELOCATION WILL NOT TAKE PLACE.

DURING RELOCATION A COMPARISON CHECK IS MADE TO INSURE NO PROGRAM LOSS

FOR FURTHER UNDERSTANDING OF HOW THE TESTS ARE PERFORMED, REFER TO THE LISTING.

10.0 CONSOLE PACKAGE ADDENDUM -----

10.1 DESCRIPTION -----

A CONSOLE PACKAGE HAS BEEN ADDED TO THIS DIAGNOSTIC TO ALLOW THE PROGRAM TO RUN WITH NO HARDWARE SWITCH REGISTER AND TO HAVE COMMUNICATIONS WITH THE DIAGNOSTIC VIA A TERMINAL. THE DIAGNOSTIC CAN BE RUN IN TWO MODES WITH THE CONSOLE PACKAGE. 1). RUNNING WITH THE CONSOLE PACKAGE ACTIVE-THIS ALLOWS THE OPERATOR CONTROL OF THE DIAGNOSTIC THROUGH THE TERMINAL. THE DIAGNOSTIC WILL ASK FOR THE VALUE OF THE PSEUDO SWITCH REGISTER BEFORE CONTINUING WITH THE EXECUTION OF THE DIAGNOSTIC. ALL ERRORS AND PASS COMPLETES WILL BE PRINTED AT THE TERMINAL. NO HALTS WILL BE EXECUTED. 2). CONSOLE PACKAGE NOT ACTIVE-THIS WILL RESULT IN THE NORMAL STANDALONE OPERATION OF THE PROGRAM AS DESCRIBED IN SECTIONS 1 THROUGH 9 OF THIS DOCUMENT.

10.2 RESTRICTIONS -----

- A. RUNNING THE CONSOLE PACKAGE REQUIRES THAT THE PSEUDO SWITCH REGISTER BE USED. HOWEVER, IF THE PROGRAM IS INITIALIZED TO USE THE HARDWARE SWITCH REGISTER THEN THE PROGRAM WILL USE THE HARDWARE SWITCH REGISTER REGARDLESS OF ANY CHANGES MADE USING THE CONSOLE TERMINAL.
- B. THE TWO SCOPE LOOPS, SECTIONS 8.1 AND A.2 OF THIS DOCUMENT, CAN NOT BE EXECUTED USING THE CONSOLE PACKAGE. TO RUN THE SCOPE LOOP SECTIONS OF THE PROGRAM, THE HARDWARE SWITCH REGISTER SHOULD BE USED.

10.3 INITIALIZATION -----

THE PROGRAM WHEN LOADED IS INITIALIZED TO USE THE HARDWARE SWITCH REGISTER AND TO RUN WITH THE CONSOLE PACKAGE NON ACTIVE. THIS WAS THE NORMAL STANDALONE CONFIGURATION OF THE PROGRAM. TO MAKE THE CONSOLE PACKAGE ACTIVE AND TO USE THE PSEUDO SWITCH REGISTER DO THE FOLLOWING:

- A. SET LOCATION 0021 BIT 0=0 TO INDICATE TO THE PROGRAM TO USE THE PSEUDO SWITCH REGISTER (LOCATION 0020)
- B. SET LOCATION 0022 BIT 3=1 TO INDICATE TO THE PROGRAM THAT THE CONSOLE PACKAGE IS ACTIVE.

STARING THE PROGRAM

- A. LOAD THE INSTRUCTION AND DATA FIELD TO THE FIELD THAT CONTAINS THE PROGRAM
- B. LOAD ADDRESS TO 0200 AND PRESS "CLEAR", THEN "RUN"
- C. THE PROGRAM WILL TYPEOUT A HEADING AND THEN PRINT SR=XXXX. XXXX IS THE VALUE OF THE SWITCH REGISTER USED. SELECT A VALUE TO USE FOR THE SWITCH REGISTER FROM THE SWITCH REGISTER TABLE IN PARAGRAPH 4.1 OF THIS DOCUMENT AND INPUT THE NUMBER USING THE CONSOLE KEYBOARD. TYPING IN FOUR DIGITS WILL CAUSE THE PROGRAM TO ECHO THE NEW VALUE OF THE SWITCH REGISTER AND START THE PROGRAM. TYPING A CARRIAGE RETURN AFTER INPUTTING A DIGIT WILL ALSO CAUSE THE PROGRAM TO ECHO THE NEW VALUE OF THE SWITCH REGISTER AND START THE PROGRAM. IF IT IS NOT DESIRED TO CHANGE THE SWITCH REGISTER VALUE, A CARRIAGE RETURN CAN BE TYPED AND THE PROGRAM WILL BE STARTED WITHOUT ECHOING THE SWITCH REGISTER VALUE.

REFER TO PARAGRAPH 4.2 OF THIS DOCUMENT FOR A DETAILED EXPLANATION OF THE SWITCH REGISTER SETTINGS.

CONTROL CHARACTERS

CONTROL G

TO GAIN CONTROL OF THE CONSOLE SWITCH REGISTER PACKAGE WHILE RUNNING THE PROGRAM, A CONTROL G MUST BE TYPED IN ON THE CONSOLE TERMINAL. WHEN CONTROL G IS TYPED THE PROGRAM WILL RESPOND BY TYPING A UP ARROW THEN G FOLLOWED BY SR=XXXX. XXXX IS THE PRESENT CONTENTS OF THE PSEUDO SWITCH REGISTER. THE OPERATOR CAN NOW CHANGE THE SETTING OF THE SWITCH REGISTER BY TYPING IN A NEW NUMBER OR NOT CHANGE IT BY TYPING IN A CARRIAGE RETURN. TYPING IN FOUR DIGITS WILL CAUSE THE PROGRAM TO ECHO THE SWITCH REGISTER VALUE TYPED AND CONTINUE THE PROGRAM USING THE VALUE TYPED AS THE NEW SWITCH REGISTER. TYPING IN ONE TO THREE DIGITS FOLLOWED BY A CARRIAGE RETURN WILL ALSO CAUSE THE PROGRAM TO ECHO THE NEW VALUE OF THE SWITCH REGISTER TYPED AND CONTINUE THE PROGRAM. TYPING ONLY A CARRIAGE RETURN WILL CAUSE THE PROGRAM TO CONTINUE WITHOUT CHANGING THE SWITCH REGISTER, AND THE PROGRAM WILL NOT ECHO THE SWITCH REGISTER VALUE. BY TYPING A LINE FEED, THE PROGRAM WILL BE RESTARTED AT ADDRESS 0200. TYPING A CONTROL C WILL CAUSE THE PROGRAM TO RETURN TO THE OPERATING SYSTEM (OS/8) AT 7600 IN FIELD 0. ILLEGAL CHARACTERS TYPED WILL RESULT IN THE CHARACTER BEING ECHOED FOLLOWED BY A QUESTION MARK AND THE SWITCH REGISTER MESSAGE BEING RETYPED.

10.5.2 CONTROL S -----

THIS IS A CONTROL CHARACTER TO STOP SENDING DATA TO A TERMINAL. IT IS USUALLY A FUNCTION OF THE TERMINAL AND IS AUTOMATICALLY SENT WHEN THE TERMINAL BUFFER IS FULL. THE BUFFER IS EMPTIED AS THE DATA IS PRINTED. AFTER ALL THE DATA IN THE TERMINAL BUFFER IS PRINTED THE TERMINAL SENDS A CONTROL Q (START SENDING DATA TO TERMINAL). THE CONTROL S IS NOT ECHOED.

BY TYPING A CONTROL S WHILE RUNNING THE DIAGNOSTIC, THE PROGRAM, WHEN THE CONTROL S IS RECOGNIZED, WILL WAIT FOR A CONTROL Q TO CONTINUE THE PROGRAM, A LINE FEED TO RESTART THE PROGRAM, OR A CONTROL C TO RETURN TO THE OPERATING SYSTEM.

10.5.3 CONTROL Q -----

THIS CONTROL CHARACTER IS TO RESUME SENDING DATA TO THE TERMINAL. IT IS USUALLY SENT AUTOMATICALLY BY A TERMINAL WHEN IT IS READY TO ACCEPT MORE DATA. THIS CONTROL CHARACTER IS NOT ECHOED.

BY TYPING A CONTROL Q AFTER A CONTROL S HAS BEEN TYPED, THE PROGRAM WILL CONTINUE FROM WHERE IT WAS INTERRUPTED FROM.

10.5.4 CONTROL C -----

THIS CONTROL CHARACTER IS USED TO RETURN CONTROL BACK TO AN OPERATING SYSTEM KEYBOARD MONITOR. THE OPERATING SYSTEM SELECTED FOR THE PDP-8 IS THE OS/8 SYSTEM WITH ITS BOOTSTRAP LOCATED IN THE LAST PAGES OF FIELD 0 AND 1. WHEN THIS CONTROL CHARACTER IS RECOGNIZED, THE PROGRAM WILL ECHO THE CONTROL CHARACTER AS AN UP ARROW THEN C, RESTORE THE LAST PAGE OF FIELD 0 AND 1, AND JUMP TO LOCATION 7600 IN FIELD 0.

10.5.5 ILLEGAL CHARACTERS -----

A CHARACTER TYPED ON THE KEYBOARD, OTHER THAN A CONTROL G, CONTROL S, OR A CONTROL C, WILL RESULT IN THE CHARACTER BEING ECHOED FOLLOWED BY A QUESTION MARK AND THE PROGRAM WILL BE CONTINUED.

10.6

END OF PASS REPORTING

THESE WILL BE AN END OF PASS MESSAGE CONTAINING THE DIAGNOSTIC NAME, THE WORD PASS AND AN OCTAL NUMBER OF PASSES. THE PROGRAM WILL ALSO TYPE THE ORIGINAL END OF PASS MESSAGE ALONG WITH THE ABOVE MESSAGE.

10.7

ERRORS

THE STANDARD ERROR REPORTS AS DESCRIBED IN SECTION 5 OF THIS DOCUMENT WILL BE USED.

10.8

SWITCH REGISTER SETTINGS

THE STANDARD SWITCH SETTINGS AS DESCRIBED IN SECTION 4 OF THIS DOCUMENT WILL BE USED. THE ONLY DIFFERENCE BEING THAT INSTEAD OF HALTING THE PROGRAM ON A SWITCH SETTING, THE PROGRAM WILL ENTER THE CONSOLE SWITCH REGISTER PACKAGE AND WAIT FOR OPERATOR RESPONSE.

10.9

FILLER CHARACTERS

IF THE CONSOLE TERMINAL REQUIRES FILLER CHARACTERS, DEPOSIT THE NUMBER OF FILLER CHARACTERS INTO LOCATION 4400 OF THE FIELD THAT THE PROGRAM IS LOCATED IN.

11.0 NON CONSOLE TERMINAL SYSTEM ADDENDUM

11.1 DESCRIPTION

THE PROGRAM HAS BEEN MODIFIED TO RUN ON THOSE SYSTEMS WITHOUT A CONSOLE TERMINAL. THIS IS DONE BY SETTING A LOCATION IN THE PROGRAM TO NON ZERO, ALL ERRORS AND FIELD LIMIT CHANGES WILL RESULT IN A HALT OR HALTS INSTEAD OF TYPEOUTS ON THE CONSOLE TERMINAL.

11.2 RESTRICTIONS

- A. FIELD LIMITS MUST BE SET AT PROGRAM START, OTHERWISE, THE PROGRAM WILL HALT TO ALLOW THE OPERATOR TO SET THE FIELD LIMITS IN THE SWITCH REGISTER.
- B. TO RUN THIS PROGRAM, A MINIMUM OF 8K OF MEMORY IS REQUIRED.
- C. MEMORIES TO BE TESTED MUST BE IN SEQUENTIAL ORDER STARTING AT FIELD 0.

11.3 INITIALIZATION

- A. SET LOCATION 0024 IN THE PROGRAM FIELD TO A NUMBER OTHER THAN 0000. SETTING THIS LOCATION TO NON-ZERO SIGNIFIES TO THE PROGRAM THAT A CONSOLE TERMINAL IS NOT AVAILABLE.
- B. THE PROGRAM WHEN LOADED IS INITIALIZED TO USE THE HARDWARE SWITCH REGISTER. IF NO HARDWARE SWITCH REGISTER IS AVAILABLE, DO THE FOLLOWING TO SELECT THE SOFTWARE PSEUDO SWITCH REGISTER (LOCATION 0020).

- 1. SET BIT 0 EQUAL TO A 0 IN LOCATION 0021 TO INDICATE TO THE PROGRAM THAT LOCATION 0020 WILL BE USED AS THE PSEUDO SWITCH REGISTER. THE PROGRAM WHEN STARTED WILL THEN SET THE PSEUDO SWITCH REGISTER TO THE FIELD LIMITS FOR A NORMAL SYSTEM STARTUP. THE PSEUDO SWITCH REGISTER WILL EQUAL XX07 WHERE XX EQUALS SWITCH REGISTER BITS PREVIOUSLY SET IN THE PSEUDO SWITCH REGISTER, 0 EQUALS STARTING FIELD LIMIT AND 7 EQUALS ENDING FIELD LIMIT.

IF IT IS DESIRED TO INITIALIZE THE FIELD LIMITS TO OTHER THAN THE ABOVE DO THE NEXT STEP.

- 2. SET LOCATION 0021 TO 00XX WHERE XX IS THE MEMORY SIZE. XX=07=8K; XX=13=12K; XX=17=16K; XX=37=32K. THE PROGRAM WHEN STARTED WILL THEN ADJUST THE PSEUDO SWITCH REGISTER TO THE APPROPRIATE FIELD LIMITS SELECTED IN LOCATION 0021.

11.4 OPERATING PROCEDURES -----

TO START THE PROGRAM:

- A. SET THE IF AND OF TO THE FIELD THAT CONTAINS THE PROGRAM.
- B. LOAD ADDRESS TO 0200
- C. IF THE PROGRAM WAS INITIALIZED TO USE THE HARDWARE SWITCH REGISTER, SET THE SWITCH REGISTER TO 0007.
- D. PRESS "INIT" AND THEN "RUN".
- E. THE PROGRAM WILL NOW RUN UNTIL AN ERROR IS ENCOUNTERED OR A SWITCH REGISTER OPTION IS SELECTED TO CAUSE THE PROGRAM TO HALT. REFER TO THE LISTING FOR ALL HALTS.
- F. SETTING THE SWITCH REGISTER TO 01XX (XX=FIELD LIMITS), WILL CAUSE THE PROGRAM TO HALT AFTER CURRENT TEST. REFER TO LISTING FOR ADDRESS OF HALT.
- G. SETTING THE SWITCH REGISTER TO 02XX (XX=FIELD LIMITS), WILL CAUSE THE PROGRAM TO HALT FOR FIELD LIMIT CHANGES VIA THE SWITCH REGISTER. REFER TO LISTING FOR ADDRESS OF HALT.

11.5 SWITCH REGISTER SETTINGS -----

- SR0=1 HALT AFTER ERROR
- SR1=1 INHIBIT ERROR HALTS EXCEPT HALT AFTER ERROR SWITCH
- SR2=1 INHIBIT OPERATION OF SR0 AND SR1
- SR3=1 INHIBIT PROGRAM RELOCATION
- SR4=1 HALT PROGRAM FOR FIELD LIMIT CHANGES VIA SR 6-11.
- SR5=1 HALT AFTER CURRENT TEST
- SR6=0 STARTING FIELD LIMIT (0-7)-NORMALLY=0
- SR9=11 ENDING FIELD LIMIT (0-7)-NORMALLY=7

11.6 ERRORS -----

ALL ERRORS ENCOUNTERED WILL RESULT IN AN ERROR HALT WITH ERROR INFORMATION IN THE AC. REFER TO THE LISTING FOR THE TYPE OF ERROR HALT AND GO TO THE APPROPRIATE PARAGRAPH BELOW. A TEST ERROR WILL RESULT IF THE CONTENTS OF A GIVEN MEMORY TEST LOCATION IS NOT EQUAL TO ITS ADDRESS OR THE COMPLEMENT OF ITS ADDRESS. A RELOCATION ERROR WILL OCCUR IF THE RELOCATION COMPARISON CHECK FAILS.

11.6.1 TEST ERROR HALTS

FOR ERRORS ENCOUNTERED TESTING MEMORY ADDRESSES, THE PROGRAM WILL HALT WITH PERTINENT INFORMATION IN THE AC. REFER TO THE STEPS BELOW FOR THE TEST ERROR INFORMATION.

- A. THE PROGRAM WILL HALT AT ADDRESS 3473 WITH THE CONTENTS OF AC BITS 9-11 EQUAL TO THE PROGRAM FIELD.
- B. PRESS "CONT". THE PROGRAM WILL HALT AT ADDRESS 3476 WITH THE CONTENTS OF THE AC EQUAL TO THE PROGRAM ADDRESS WHERE THE ERROR JMS OCCURRED.
- C. PRESS "CONT". THE PROGRAM WILL HALT AT ADDRESS 3502 WITH THE CONTENTS OF AC BITS 9-11 EQUAL TO THE FIELD CONTAINING THE ERROR.
- D. PRESS "CONT". THE PROGRAM WILL HALT AT ADDRESS 3505 WITH THE CONTENTS OF THE AC EQUAL TO THE ADDRESS OF LOCATION IN ERROR.
- E. PRESS "CONT". THE PROGRAM WILL HALT AT ADDRESS 3510 WITH THE CONTENTS OF THE AC EQUAL TO WHAT THE DATA SHOULD BE.
- F. PRESS "CONT". THE PROGRAM WILL HALT AT ADDRESS 3513 WITH THE CONTENTS OF THE AC EQUAL TO WHAT THE DATA WAS.
- G. PRESS "CONT". THE PROGRAM WILL HALT AT ADDRESS 3522 WITH THE CONTENTS OF THE AC EQUAL TO A TEST NUMBER (1-4) OF THE TEST RUNNING WHEN THE FAILURE OCCURRED.
- H. PRESS "CONT" TO CONTINUE THE PROGRAM ON TO THE NEXT SEQUENTIAL TEST MEMORY ADDRESS.
- I. TEST ERROR HALTS MAY BE INHIBITED FROM HALTING BY SETTING SR1 TO A 1.

11.6.2 RELOCATION ERROR HALTS -----

ALL RELOCATION ERRORS WILL RESULT IN A HALT WITH PERTINENT INFORMATION IN THE AC. REFER TO THE STEPS BELOW FOR THE ERROR INFORMATION.

- A. THE PROGRAM WILL HALT AT ADDRESS 3527 WITH THE CONTENTS OF AC BITS 9-11 EQUAL TO THE FIELD THE PROGRAM IS LOCATED IN.
- B. PRESS "CONT". THE PROGRAM WILL HALT AT ADDRESS 3532 WITH THE CONTENTS OF THE AC EQUAL TO THE ADDRESS WHERE THE ERROR JMS OCCURED.
- C. PRESS "CONT". THE PROGRAM WILL HALT AT ADDRESS 3536 WITH THE CONTENTS OF AC BITS 9-11 EQUAL TO THE FIELD THE PROGRAM IS RELOCATING TO.
- D. PRESS "CONT". THE PROGRAM WILL HALT AT ADDRESS 3541 WITH THE CONTENTS OF THE AC EQUAL TO THE ADDRESS OF LOCATION IN ERROR.
- E. PRESSING "CONT" AGAIN WILL RESULT IN THE PROGRAM CONTINUING WITH THE NEXT SEQUENTIAL MEMORY LOCATION. CONTINUING IS NOT ADVISABLE BECAUSE THE PROGRAM MAY HAVE BEEN CHANGED DURING RELOCATION.

11.6.3 HALT AFTER ERROR -----

THE PROGRAM WILL ONLY HALT HERE IF SWITCH REGISTER 0 EQUALS A ONE AND AN ERROR OCCURED. IF SWITCH REGISTER TWO EQUALS A ONE, NO HALT WILL OCCUR.

- A. THE PROGRAM WILL HALT AT ADDRESS 2262 WITH THE CONTENTS OF THE AC EQUAL TO THE PROGRAM LOCATION WHERE THE ERROR JMS LOCATION.
- B. PRESSING "CONT" WILL CAUSE THE PROGRAM TO CONTINUE FROM THE POINT WHERE THE ERROR WAS DETECTED.

APT NOTES

SEQ 0017

ALL OF THE FOLLOWING NOTES APPLY ONLY WHEN THE PROGRAM IS BEING RUN ON AN APT SYSTEM.

1. FOR MORE INFORMATION SEE THE FOLLOWING DOCUMENTS.
 - A. STANDARD APT SYSTEM TO PDPS DIAGNOSTIC INTERFACE.
 - B. APT SYSTEM MANAGERS GUIDE.
2. FOR ANY DIFFERENCES BETWEEN THESE NOTES AND THE REST OF THE DOCUMENT, THESE NOTES WILL PREVAIL. (SEE THE LISTING ALSO.)
3. ALL CODE THAT HAS BEEN ADDED (INSERTED) FOR APT WILL CONTAIN THE EXPRESSION: /APT/

ANY ORIGINAL CODE NEGATED FOR APT HAS BEEN 'REMOVED' BY PRECEDING IT WITH THE EXPRESSION: /*APT*/
4. IF BIT 0 OF HCW1 IS A '1' THEN THE HARDWARE SWITCH REGISTER WILL BE USED, REGARDLESS OF LOAD METHOD (SCRIPT OR DUMP).

THE FOLLOWING NOTES APPLY ONLY WHEN THE LOAD METHOD WAS SCRIPT LOAD.

5. SWITCHES:

SR3, SR6=0 & SR9=11 ARE USED AS DESCRIBED IN THE DOCUMENT.

SR6=0 & SR9=11 MAY BE USED TO SPECIFY FIELD LIMITS INSTEAD OF HCW1. (WILL BE USED IF HCW1 BITS 7-11 = 0.)

FIELD 7 CANNOT BE SPECIFIED.
6. ERRORS:

ALL ERRORS CALL APT.

LISTING

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/KM8E EXTENDED MEMORY ADDRESS TEST MAINDEC=08-DHKMC-B=L
/EXTENDED ADDRESS TEST FOR KM8-E EXTENDED MEMORY (VER B)
/COPYRIGHT 1971, 1973, 1976, DIGITAL EQUIPMENT CORPORATION, MAYNARD, MASS. 01754
/PROGRAMMER, VERNON FREY

/      MODIFIED FOR APT      APRIL 1975      D. MACOMBER
/      MODIFIED TO RUN WITH NO CONSOLE - APRIL 1976 - B. HANSEN
/      MODIFIED FOR A NEW CONSOLE PACKAGE - APRIL 1976 - B. HANSEN

/
/
/8W0=1      HALT AFTER ERROR
/8W1=1      INHIBIT ERROR TYPEOUT
/8W2=1      BELL ON ERROR (USEFUL FOR MAINTENANCE)
/8W3=1      INHIBIT PROGRAM RELOCATION
/8W4=1      CHANGE STACK LIMITS
/8W5=1      HALT AFTER CURRENT TEST
/8W6=8W8    STARTING STACK LIMIT (0-7)
/8W9=8W11   ENDING STACK LIMIT (0-7)

/
/
/PROGRAM STARTING ADDRESS
/0200      RUN ALL TESTS
/0201      RUN ONLY TEST 1
/0202      RUN ONLY TEST 2
/0203      RUN ONLY TEST 3
/0204      RUN ONLY TEST 4

/
/
/
/IDT COMMANDS FOR THE MC8-E EXTENDED MEMORY & INTERRUPT

6004      GTF=6004      /GET INTERRUPT FLAGS
                        /AC0 LINK
                        /AC1 GREATER THAN FLAG
                        /AC2 INTERRUPT BUS
                        /AC3 INTERRUPT INHIBIT FLIP-FLOP
                        /AC4 INTERRUPT ON
                        /AC5 USER FLAG
                        /AC6=8 INSTRUCTION FIELD
                        /AC9=11 DATA FIELD
6005      RTF=6005      /RESTORE INTERRUPT FLAGS
                        /AC0 LINK
                        /AC1 GREATER THAN FLAG
                        /I INTERRUPT INHIBIT FLIP-FLOP
                        /I INTERRUPT ON
                        /AC5 USER FLAG
                        /AC6=8 INSTRUCTION BUFFER
                        /AC9=11 DATA FIELD
6201      CDF0=6201     /CHANGE TO DATA FIELD 0
6211      CDF1=6211     /CHANGE TO DATA FIELD 1

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6221      CDF2=6221     /CHANGE TO DATA FIELD 2
6231      CDF3=6231     /CHANGE TO DATA FIELD 3
6241      CDF4=6241     /CHANGE TO DATA FIELD 4
6251      CDF5=6251     /CHANGE TO DATA FIELD 5
6261      CDF6=6261     /CHANGE TO DATA FIELD 6
6271      CDF7=6271     /CHANGE TO DATA FIELD 7
6202      CIF0=6202     /CHANGE TO INSTRUCTION FIELD 0
6212      CIF1=6212     /CHANGE TO INSTRUCTION FIELD 1
6222      CIF2=6222     /CHANGE TO INSTRUCTION FIELD 2
6232      CIF3=6232     /CHANGE TO INSTRUCTION FIELD 3
6242      CIF4=6242     /CHANGE TO INSTRUCTION FIELD 4
6252      CIF5=6252     /CHANGE TO INSTRUCTION FIELD 5
6262      CIF6=6262     /CHANGE TO INSTRUCTION FIELD 6
6272      CIF7=6272     /CHANGE TO INSTRUCTION FIELD 7
6203      CBF0=6203     /CHANGE TO DATA AND INSTRUCTION FIELD 0
6213      CBF1=6213     /CHANGE TO DATA AND INSTRUCTION FIELD 1
6223      CBF2=6223     /CHANGE TO DATA AND INSTRUCTION FIELD 2
6233      CBF3=6233     /CHANGE TO DATA AND INSTRUCTION FIELD 3
6243      CBF4=6243     /CHANGE TO DATA AND INSTRUCTION FIELD 4
6253      CBF5=6253     /CHANGE TO DATA AND INSTRUCTION FIELD 5
6263      CBF6=6263     /CHANGE TO DATA AND INSTRUCTION FIELD 6
6273      CBF7=6273     /CHANGE TO DATA AND INSTRUCTION FIELD 7
6204      CINT=6204     /CLEAR USER INTERRUPT (TIME SHARE)
6214      RDF=6214      /READ DATA FIELD INTO AC BITS 6-8
6224      RIF=6224      /READ INSTRUCTION FIELD INTO AC BITS 6-8
6234      RIB=6234      /READ INTERRUPT BUFFER
                        /AC6=8 INSTRUCTION FIELD IN USE BEFORE LAST
                        / PROGRAM INTERRUPT.
                        /
                        /AC9=11 DATA FIELD IN USE BEFORE LAST
                        / PROGRAM INTERRUPT.
                        /
6244      RMF=6244      /RESTORE MEMORY FIELD
                        /INSTRUCTION FIELD LOADED FROM SAVE FIELD 0-2
                        /DATA FIELD LOADED FROM SAVE FIELD 3-5
6254      SINT=6254     /SKIP ON USER INTERRUPT (TIME SHARE)
6264      CUF=6264      /CLEAR USER FLAG (TIME SHARE)
6274      SUF=6274      /SET USER FLAG (TIME SHARE)

0020      *20

0020      0000      PSR, 0      /APT/
0021      4000      MCW1, 4000 /APT/
0022      0000      MCW2, 0     /APT/

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4425      LISN=JMS I      XLISP
4426      PRNTH8=JMS I     MESAGP
4427      ONEOCT=JMS I     ONEOCP
4430      TWOOCT=JMS I     TWOOCP
4432      PRNT1=JMS I      XPRN1P
4431      FOROCT=JMS I     FOROCP
4433      PRNT2=JMS I      XPRN2P
4434      PRNT4=JMS I      XPRN4P

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4435 SPACE2=JMS I SPCX2P
4436 TYPE=JMS I TYPEP
4437 CRLF=JMS I CRLFP
4440 GETSR= JMS I GETSRX
4441 CHKCON= JMS I CHKCAC

4440 LAS= GETSR

0024 *24

0024 0000 NOTTY, 0 /THIS FLAG SET TO NON ZERO SIGNIFIES THAT
/NO CONSOLE TERMINAL IS AVAILABLE

0025 4475 XLISP, XLISN
0026 4600 MESAGP, MESAGX
0027 4401 ONEOCP, ONEOCK
0030 4411 TWOOCK, TWOOCK
0031 4640 FORDCP, FORDCK
0032 4667 XPRN1P, XPRNT1
0033 4424 XPRN2P, XPRNT2
0034 4654 XPRN4P, XPRNT4
0035 4467 SPCX2P, SPACK2
0036 4435 TYPEP, XTYPE
0037 4452 CRLFP, XCRLF
0040 4240 GETSRX, SRGET
0041 4562 CHKCAC, CONCHK

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/
/CONSTANTS AND POINTERS
/
0042 4000 SW0, 4000 /HALT AFTER ERROR
0043 2000 SW1, 2000 /INHIBIT ERROR TYPEOUT
0044 1000 SW2, 1000 /REFL ON ERROR
0045 0400 SW3, 400 /INHIBIT PROGRAM RELOCATION
0046 0200 SW4, 200 /CHANGE STACK LIMITS
0047 0100 SW5, 100 /HALT AFTER CURRENT TEST

0050 0000 STACK0, 0 /
0051 0000 STACK1, 0 /
0052 0000 STACK2, 0 /
0053 0000 STACK3, 0 /STACKS CONTAIN 0 IF SELECTED FOR TESTING
0054 0000 STACK4, 0 /
0055 0000 STACK5, 0 /
0056 0000 STACK6, 0 /
0057 0000 STACK7, 0 /
0060 0000 STK0, 0 /
0061 0000 STK1, 0 /
0062 0000 STK2, 0 /
0063 0000 STK3, 0 /0 IF RELOCATE
0064 0000 STK4, 0 /
0065 0000 STK5, 0 /
0066 0000 STK6, 0 /

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0067 0000 STK7, 0 /
0070 0000 NORELO, 0 /PRG RELOCATION CONTROL (0=INH)
0071 1715 KABOVE, ABOVE /CONTROL UPPER STACKS NOT TESTED
0072 1734 KBELOW, BELOW /CONTROL LOWER STACKS NOT TESTED
0073 0000 HEAD0, 0 /ERROR HEADING CONTROL
0074 0000 INAME, 0 /PRG IN SEL STACK
0075 0000 LEGAL0, 0 /LEGAL STACK SELECTION
0076 0000 RUNITST, 0 /6003=ALL, 0001=1, 0002=2, 2000=3, 4000=4
0077 0000 TESTAD, 0 /TEST ADDRESS COUNTER
0100 0000 KBINT, 0 /HIGHEST ACTUAL STACK IN SYSTEM
0101 0000 SSL, 0 /STARTING STACK LIMIT 0PX0
0102 0000 ESL, 0 /ENDING STACK LIMIT 000X
0103 0000 STKPIN, 0 /STACK PRG IS IN 00X0
0104 0000 STKYST, 0 /STACK SEL FOR TEST 00X0
0105 0000 BDATA, 0 /BAD DATA
0106 0000 GDATA, 0 /GOOD DATA
0107 0000 MOVE, 0 /RELOCATION ADDRESS
0110 1745 KDOWN, DOWN /CONTROL LOWER STACKS TESTED
0111 0000 TEMP, 0 /INDIRECT ADDRESS TEMP STORAGE = CHEXN
0112 0000 COUNT, 0 /CHECKERBOARD ERROR COUNTER
0113 0000 ERRLOC, 0 /CODERR
0114 7777 M1, -1 /CODERR = TEST 3 & 4
0115 7776 M2, -2 /MESSAGE = LEGAL
0116 7775 M3, -3 /MESSAGE
0117 7774 M4, -4 /MESSAGE = 4 WORDS
0120 7744 M34, -34 /MESSAGE

SW911, /CB/ENDING STACK LIMIT (0-7).

0121 0007 K7, 7 /CODERR = ERRC = STACKS
0122 0010 K10, 10 /CHEXN
0123 0020 K20, 20 /CHEXN
0124 0030 K30, 30 /CHEXN
0125 0040 K40, 40 /CHEXN
0126 0050 K50, 50 /CHEXN
0127 0060 K60, 60 /CHEXN

SW60, /CB/STARTING STACK LIMIT (0-7).

0130 0070 K70, 70 /CHEXN
0131 0077 K77, 77 /SIXTY - MESSAGE
0132 0207 K207, 207 /MESSAGE = CODERR
0133 0212 K212, 212 /MESSAGE
0134 0215 K215, 215 /MESSAGE
0135 0240 K240, 240 /TOSSEL
0136 4060 K4060, 4060 /CODERR = ERRC
0137 6201 K6201, 6201 /CDF 0
0140 2042 XMSGAG, MESSAGE /TTY ROUTINE POINTER
0141 2000 XSIXTY, SIXTY /SIXTY ROUTINE POINTER
0142 2201 XCODER, CODERR /ERROR ROUTINE POINTER
0143 2200 XRETUR, RETURN /ERROR RETURN POINTER
0144 2231 XSTOP, STOP /STOP ROUTINE POINTER
0145 2250 XADDER, ADDER /ADDRESS OF ERROR TYPEOUT POINTER
0146 0000 MINS, 0 /FIVE MINUTE CONTROL

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0000 0000 *0
0000 0000 0
0001 5001 JMP
0002 0002 2
0003 0003 3

0004 4334 KSFCMK, CMKKSF
0005 4026 IAPTER, APTER /APT/
0006 4000 IAPTOK, APTOK /APT/

0200 *200
/
/KM8-E EXTENDED MEMORY ADDRESS TEST (EABE)
/

0200 5777' START, JMP APTIZ /SETUP FOR APT/SETUP FOR NO TTY/SETUP FOR CONSOLE
0201 5776' JMP RUN1 /TEST 1
0202 5775' JMP RUN2 /TEST 2
0203 5774' JMP RUN3 /TEST 3
0204 5773' JMP RUN4 /TEST 4
0205 3076 EXTAD0, DCA RUNTST /TEST CONTROL
0206 6002 IOF
0207 6224 RIF
0210 1137 TAD K6201
0211 3212 DCA *1
0212 6201 CDF 0 /MAKE DATA FIELD=INST FIELD

/APT/ IF UNDER APT CONTROL NEXT TWO LOCS WILL = NOP.

APTN00, /APT/

0213 4772' JMS TITLE /TYPEOUT PROGRAM TITLE

APTN01, /APT/

0214 4771' CHEXA, JMS SETSW /TYPEOUT TO SETUP SWITCHES
0215 7240 STA
0216 3070 DCA NORELO /CLEAR INH RELOCATION
0217 3077 DCA TESTAD /CLEAR TEST ADDR COUNTER
0220 7240 STA
0221 3073 DCA HEAD1 /RESET ERROR HEADING
0222 1146 TAD MINS
0223 3770' DCA FIVE /SETUP COUNTER
0224 4767' JMS DOWN+2 /CLEAR STACK SELECTION CONTROLS
0225 4440 LAS
0226 0130 AND SW6A
0227 3101 DCA SSL /STARTING STACK LIMIT
0230 4440 LAS
0231 0121 AND SW911
0232 3102 DCA ESL /ENDING STACK LIMIT
0233 4766' JMS MSSL /OBTAIN -SSL IN AC BITS 9-11
0234 1102 TAD ESL
0235 7640 SZA CLA /SKIP IF SSL EQUALS ESL
0236 5262 JMP CHEXC /CONTINUE CHECK
0237 6224 RIF /READ INSTRUCTION FIELD

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0240 7041 CIA
0241 1101 TAD SSL
0242 7650 SNA CLA
0243 5765' JMP PINF /PROGRAM IS IN THE SELECTED FIELD
0244 3070 DCA NORELO /INHIBIT PROGRAM RELOCATION
0245 1364 TAD (CHEXB
0246 3763' DCA ABOVE+1 /STORE RETURN ADDRESS
0247 1102 TAD ESL
0250 1071 TAD KAROVE
0251 3111 DCA TEMP
0252 5511 JMP I TEMP /INCREMENT UPPER FIELDS NOT TESTED
0253 1362 CHEXB, TAD (CHEXE
0254 3761' DCA BELOW+1 /STORE RETURN ADDRESS
0255 1102 TAD ESL
0256 7041 CIA
0257 1072 TAD KRELOW
0260 3111 DCA TEMP
0261 5511 JMP I TEMP /INCREMENT LOWER FIELDS NOT TESTED
0262 4766' CHEXC, JMS MSSL /OBTAIN -SSL IN AC BITS 9-11
0263 1102 TAD ESL
0264 7710 SPA CLA
0265 5302 JMP CHEXD /STARTING FIELD IS GREATER THAN ENDING FIELD
0266 1360 TAD (CHEXC1
0267 3763' DCA ABOVE+1 /STORE RETURN ADDRESS
0270 1102 TAD ESL
0271 1071 TAD KAROVE
0272 3111 DCA TEMP
0273 5511 JMP I TEMP /INCREMENT UPPER FIELDS NOT TESTED
0274 1362 CHEXC1, TAD (CHEXE
0275 3761' DCA BELOW+1 /STORE RETURN ADDRESS
0276 4766' JMS MSSL /OBTAIN -SSL IN AC BITS 9-11
0277 1072 TAD KRELOW
0300 3111 DCA TEMP
0301 5511 JMP I TEMP /INCREMENT LOWER FIELDS NOT TESTED
0302 1357 CHEXD, TAD (CHEXD1
0303 3761' DCA BELOW+1 /STORE RETURN ADDRESS
0304 4766' JMS MSSL /OBTAIN -SSL IN AC BITS 9-11
0305 1072 TAD KRELOW
0306 3111 DCA TEMP
0307 5511 JMP I TEMP /INCREMENT ALL LOWER FIELDS
0310 1362 CHEXD1, TAD (CHEXE
0311 3767' DCA DOWN+2 /STORE RETURN ADDRESS
0312 1102 TAD ESL
0313 7041 CIA
0314 1110 TAD KDOWN
0315 3111 DCA TEMP
0316 5511 JMP I TEMP /RESTORE LOWER FIELDS TESTED
0317 4756' CHEXE, JMS HIGHEST /FIND SYSTEMS HIGHEST STACK
0320 1100 TAD KBINT
0321 1355 TAD (200
0322 3100 DCA KBINT /MAKE HIGHEST STACK 0-7 FOR TYPEOUT

/APT/ IF UNDER APT CONTROL NEXT LOC WILL = NOP.

APTN02, /APT/

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0323 4754* JMS TSTSYS /TYPEOUT # OF STACKS IN SYSTEM
0324 1353 TAD CMEXE2
0325 3763* DCA ABOVE-1 /STORE RETURN ADDRESS
0326 1100 TAD KBINT
0327 0121 AND K7
0330 1071 TAD KABOVE
0331 3111 DCA TEMP
0332 5511 JMP I TEMP /INCREMENT UPPER STACKS NOT IN SYSTEM
0333 0400
0334 3017
0335 0260
0336 2731
0337 0310
0360 0274
0361 1735
0362 0317
0363 1714
0364 0253
0365 2127
0366 1751
0367 1747
0370 1262
0371 2624
0372 2600
0373 1624
0374 1617
0375 1612
0376 1605
0377 3600
0400

```

PAGE

/APT/ IF UNDER APT CONTROL NEXT LOC WILL = NOP.

APTN05,

/APT/

```

0400 4777* CMEXE2, JMS TOSEL /TYPEOUT STACK TEST SELECTION
0401 4210 JMS LEGAL
0402 1070 TAD NORELO
0403 7650 SNA CLA
0404 5276 JMP CMEXM
0405 4776* JMS CMKSW3 /CHECK PROG RELO SW
0406 5775* JMP CMEXO /RELOCATE PROGRAM
0407 5332 JMP CMEXN /INHIBIT PROGRAM RELOCATION

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/CHECK FOR LEGAL STACK SELECTION

```

0410 0000 /LEGAL, 0
0411 7300 CLA CLL
0412 3074 DCA INSAME /CLEAR SAME CONTROL
0413 1115 TAD M2
0414 3075 DCA LEGAL0 /SETUP LEGAL CONTROL
0415 3104 DCA STKTST

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0416 1050 TAD STACK0
0417 4263 JMS LEGAL0
0420 1122 TAD K10
0421 3104 DCA STKTST
0422 1051 TAD STACK1
0423 4263 JMS LEGAL0
0424 1123 TAD K20
0425 3104 DCA STKTST
0426 1052 TAD STACK2
0427 4263 JMS LEGAL0
0430 1124 TAD K30
0431 3104 DCA STKTST
0432 1053 TAD STACK3
0433 4263 JMS LEGAL0
0434 1125 TAD K40
0435 3104 DCA STKTST
0436 1054 TAD STACK4
0437 4263 JMS LEGAL0
0440 1126 TAD K50
0441 3104 DCA STKTST
0442 1055 TAD STACK5
0443 4263 JMS LEGAL0
0444 1127 TAD K60
0445 3104 DCA STKTST
0446 1056 TAD STACK6
0447 4263 JMS LEGAL0
0450 1130 TAD K70
0451 3104 DCA STKTST
0452 1057 TAD STACK7
0453 4263 JMS LEGAL0
0454 2075 ISZ LEGAL0
0455 5774* JMP NOSTK /NO STACK SELECTION
0456 1074 TAD INSAME
0457 7640 SZA CLA
0460 5773* JMP PINF /PROG IN SELECTED FIELD
0461 3070 DCA NORELO /ONLY 1 STACK SELECTED
0462 5610 JMP I LEGAL
/LEGAL STACK SELECTION SUBROUTINE
/LEGAL, 0
0463 0000 SZA CLA
0464 7640 JMP I LEGAL0 /NOT SELECTED
0465 5663 ISZ LEGAL0
0466 2075 SKP
0467 7410 JMP I LEGAL
0470 5610 RIF
0471 6224 DCA STKPIN
0472 3103 JMS SAME /PROG IN SEL STACK
0473 4772* ISZ INSAME /YES
0474 2074 JMP I LEGAL0
0475 5663

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/NO PROGRAM RELOCATION AND TEST ONLY 1 STACK

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0476 6224 CHEXM, RIF
0477 3103 DCA STKPIN /STACK PROGRAM IS IN
0500 1371 TAD (STACK0-1
0501 3017 DCA 17
0502 3111 DCA TEMP
0503 1417 CHEXM1, TAD 1 17 /FIND STACK SEL FOR TEST
0504 7650 SNA CLA
0505 5310 JMP CHEXM2
0506 2111 ISZ TEMP
0507 5303 JMP CHEXM1
0510 1111 CHEXM2, TAD TEMP
0511 7104 CLL RAL
0512 7006 RTL
0513 3104 DCA STKYST /STACK SEL FOR TEST

/IF UNDER APT NEXT LOCATION WILL = NOP

APTN06,

0514 4770' JMS PNOREL /TYPEOUT NO RELOCATION
0515 4767' CHEXM3, JMS TEST /TEST THE SELECTED STACK

/APT/ IF UNDER APT CONTROL NEXT LOC WILL = JMP APTJ50.

APTJ00, /APT/

0516 4440 LAS /HALT AFTER TEST
0517 0047 AND SW5
0520 7650 SNA CLA
0521 5325 JMP .+4 /DO NOT HALT AFTER TEST
0522 4441 CHKCON /CHECK TO SEE IF CONSOLE ACTIVE
0523 7402 HLT /CONSOLE INACTIVE=HALT AFTER TEST
0524 4766' JMS PSEUDO /HALT AFTER TEST-ASK SR QUESTION
0525 4440 LAS
0526 0046 AND SW4 /CHANGE STACK LIMITS?
0527 7640 SZA CLA
0530 5765' JMP CHEXA /YES

APTJ50, /APT/

0531 5315 JMP CHEXM3 /NO

/
/NO PROGRAM RELOCATION BUT TEST ALL SELECTED STACKS
/

/APT/ IF UNDER APT CONTROL NEXT LOC WILL = NOP.

APTN03, /APT/

0532 4770' CHEXM, JMS PNOREL /TYPEOUT NO RELOCATION
0533 4764' CHEXM0, JMS CHEXM1 /TEST SEL'D STACKS

/APT/ IF UNDER APT CONTROL NEXT LOC WILL = JMP APTJ51.

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APTJ01, /APT/

0534 4440 LAS /HALT AFTER TEST
0535 0047 AND SW5
0536 7650 SNA CLA
0537 5343 JMP .+4 /
/NO=DO NOT HALT AFTER TEST
0540 4441 CHKCON /CHECK TO SEE IF CONSOLE ACTIVE
0541 7402 HLT /INACTIVE CONSOLE=HALT AFTER TEST
0542 4766' JMS PSEUDO /ASK SR QUESTION IF CONSOLE ACTIVE
0543 4440 LAS
0544 0046 AND SW4 /CHANGE STACK LIMITS?
0545 7640 SZA CLA
0546 5765' JMP CHEXA /YES

APTJ51, /APT/

0547 4776' JMS CHKSW3 /NO
0550 5775' JMP CHEX0 /RELOCATE
0551 5333 JMP CHEXM0 /CONTINUE

0564 0600
0565 0214
0566 4247
0567 1202
0570 2647
0571 0047
0572 1640
0573 2127
0574 2152
0575 1000
0576 1647
0577 3077
0600 PAGE

/
/TEST STACKS CONTROL
/
0600 0000 CHEXM1, 0
0601 7200 CLA
0602 6224 RIF
0603 3103 DCA STKPIN /STACK PROGRAM IS IN
0604 1057 TAD STACK7
0605 7640 SZA CLA
0606 5222 JMP CHEXM2
0607 1130 TAD K70
0610 3104 DCA STKYST /STACK SEL FOR TEST
0611 3112 DCA COUNT
0612 4777' JMS SAME /PROG IN SEL STACK?
0613 5222 JMP CHEXM2 /YES
0614 4776' JMS TEST /NO = TEST THE SEL STACK
0615 1112 TAD COUNT
0616 7640 SZA CLA
0617 2067 ISZ
0620 7410 SKP
0621 5217 JMP .-2

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0622	1056	CHEXN2, TAD	STACK6	
0623	7640	SZA CLA		
0624	5240	JMP	CHEXN3	
0625	1127	TAD	K60	
0626	3104	DCA	STKTST	/STACK SEL FOR TEST
0627	3112	DCA	COUNT	
0630	4777	JMS	SAME	/PROG IN SEL STACK?
0631	5240	JMP	CHEXN3	/YES
0632	4776	JMS	TEST	/NO - TEST THE SEL STACK
0633	1112	TAD	COUNT	
0634	7640	SZA CLA		
0635	2066	ISZ	STK6	
0636	7410	SKP		
0637	5235	JMP	.-2	
0640	1055	CHEXN3, TAD	STACK5	
0641	7640	SZA CLA		
0642	5256	JMP	CHEXN4	
0643	1126	TAD	K50	
0644	3104	DCA	STKTST	/STACK SEL FOR TEST
0645	3112	DCA	COUNT	
0646	4777	JMS	SAME	/PROG IN SEL STACK?
0647	5256	JMP	CHEXN4	/YES
0650	4776	JMS	TEST	/NO - TEST THE SEL STACK
0651	1112	TAD	COUNT	
0652	7640	SZA CLA		
0653	2065	ISZ	STK5	
0654	7410	SKP		
0655	5253	JMP	.-2	
0656	1054	CHEXN4, TAD	STACK4	
0657	7640	SZA CLA		
0660	5274	JMP	CHEXN5	
0661	1125	TAD	K40	
0662	3104	DCA	STKTST	/STACK SEL FOR TEST
0663	3112	DCA	COUNT	
0664	4777	JMS	SAME	/PROG IN SEL STACK?
0665	5274	JMP	CHEXN5	/YES
0666	4776	JMS	TEST	/NO - TEST THE SEL STACK
0667	1112	TAD	COUNT	
0670	7640	SZA CLA		
0671	2064	ISZ	STK4	
0672	7410	SKP		
0673	5271	JMP	.-2	
0674	1053	CHEXN5, TAD	STACK3	
0675	7640	SZA CLA		
0676	5312	JMP	CHEXN6	
0677	1124	TAD	K30	
0700	3104	DCA	STKTST	/STACK SEL FOR TEST
0701	3112	DCA	COUNT	
0702	4777	JMS	SAME	/PROG IN SEL STACK?
0703	5312	JMP	CHEXN6	/YES
0704	4776	JMS	TEST	/NO - TEST THE SEL STACK
0705	1112	TAD	COUNT	
0706	7640	SZA CLA		
0707	2063	ISZ	STK3	
0710	7410	SKP		

0711	5307	JMP	.-2	
0712	1052	CHEXN6, TAD	STACK2	
0713	7640	SZA CLA		
0714	5330	JMP	CHEXN7	
0715	1123	TAD	K20	
0716	3104	DCA	STKTST	/STACK SEL FOR TEST
0717	3112	DCA	COUNT	
0720	4777	JMS	SAME	/PROG IN SEL STACK?
0721	5330	JMP	CHEXN7	/YES
0722	4776	JMS	TEST	/NO - TEST THE SEL STACK
0723	1112	TAD	COUNT	
0724	7640	SZA CLA		
0725	2062	ISZ	STK2	
0726	7410	SKP		
0727	5325	JMP	.-2	
0730	1051	CHEXN7, TAD	STACK1	
0731	7640	SZA CLA		
0732	5346	JMP	CHEXN8	
0733	1122	TAD	K10	
0734	3104	DCA	STKTST	/STACK SEL FOR TEST
0735	3112	DCA	COUNT	
0736	4777	JMS	SAME	/PROG IN SEL STACK?
0737	5346	JMP	CHEXN8	/YES
0740	4776	JMS	TEST	/NO - TEST THE SEL STACK
0741	1112	TAD	COUNT	
0742	7640	SZA CLA		
0743	2061	ISZ	STK1	
0744	7410	SKP		
0745	5343	JMP	.-2	
0746	1050	CHEXN8, TAD	STACK0	
0747	7640	SZA CLA		
0750	5361	JMP	CHEXN9	
0751	3104	DCA	STKTST	/STACK SEL FOR TEST
0752	3112	DCA	COUNT	
0753	4777	JMS	SAME	/PROG IN SEL STACK?
0754	5361	JMP	CHEXN9	/YES
0755	4776	JMS	TEST	/NO - TEST THE SEL STACK
0756	1112	TAD	COUNT	
0757	7640	SZA CLA		
0760	2060	ISZ	STK0	
0761	5600	CHEXN9, JMP I	CHEXN1	
0762	5360	JMP	.-2	
0776	1200			
0777	1640			
	1000			

PAGE

/

/CHECK ALL SELECTED STACKS FROM EACH SELECTED STACK

/

/APT/ IF UNDER APT CONTROL NEXT LOC WILL = NOP.

APTN04, /APT/

```
1000 4777' CHEX0, JMS PREL /TYPEOUT RELOCATION
1001 4776' JMS RESTK /RESTORE STK(3)
1002 4775' JMS CHEXN1 /TEST FROM PRESENT STACK
1003 4774' JMS CHKSW3
1004 7410 SKP
1005 5773' JMP CHEXN
1006 6224 CHEX0A, RIF
1007 3103 DCA STKPIN /STACK PROGRAM IS IN
1010 1067 TAD STK7
1011 7640 SZA CLA
1012 5224 JMP CHEX00
1013 1130 TAD K70
1014 3104 DCA STKTST
1015 4772' JMS SAME /STACK SEL FOR MOVE TO
1016 7410 SKP /PROG IN MOVE STACK?
1017 4771' JMS RELO /YES
1020 4775' JMS CHEXN1 /NO - RELOCATE PROGRAM
1021 4774' JMS CHKSW3 /TEST ALL SEL STACKS
1022 7410 SKP
1023 5773' JMP CHEXN
1024 1066 CHEX00, TAD STK6
1025 7640 SZA CLA
1026 5240 JMP CHEX01
1027 1127 TAD K60
1030 3104 DCA STKTST
1031 4772' JMS SAME /STACK SEL FOR MOVE TO
1032 7410 SKP /PROG IN MOVE STACK?
1033 4771' JMS RELO /YES
1034 4775' JMS CHEXN1 /NO - RELOCATE PROGRAM
1035 4774' JMS CHKSW3 /TEST ALL SEL STACKS
1036 7410 SKP
1037 5773' JMP CHEXN
1040 1065 CHEX01, TAD STK5
1041 7640 SZA CLA
1042 5254 JMP CHEX02
1043 1126 TAD K50
1044 3104 DCA STKTST
1045 4772' JMS SAME
1046 7410 SKP
1047 4771' JMS RELO
1050 4775' JMS CHEXN1
1051 4774' JMS CHKSW3
1052 7410 SKP
1053 5773' JMP CHEXN
1054 1064 CHEX02, TAD STK4
1055 7640 SZA CLA
1056 5270 JMP CHEX03
1057 1125 TAD K40
1060 3104 DCA STKTST
1061 4772' JMS SAME
1062 7410 SKP
1063 4771' JMS RELO
1064 4775' JMS CHEXN1
1065 4774' JMS CHKSW3
```

```
1066 7410 SKP
1067 5773' JMP CHEXN
1070 1063 CHEX03, TAD STK3
1071 7640 SZA CLA
1072 5380 JMP CHEX04
1073 1124 TAD K30
1074 3104 DCA STKTST
1075 4772' JMS SAME
1076 7410 SKP
1077 4771' JMS RELO
1100 4775' JMS CHEXN1
1101 4774' JMS CHKSW3
1102 7410 SKP
1103 5773' JMP CHEXN
1104 1062 CHEX04, TAD STK2
1105 7640 SZA CLA
1106 5320 JMP CHEX05
1107 1123 TAD K20
1110 3104 DCA STKTST
1111 4772' JMS SAME
1112 7410 SKP
1113 4771' JMS RELO
1114 4775' JMS CHEXN1
1115 4774' JMS CHKSW3
1116 7410 SKP
1117 5773' JMP CHEXN
1120 1061 CHEX05, TAD STK1
1121 7640 SZA CLA
1122 5334 JMP CHEX06
1123 1122 TAD K10
1124 3104 DCA STKTST
1125 4772' JMS SAME
1126 7410 SKP
1127 4771' JMS RELO
1130 4775' JMS CHEXN1
1131 4774' JMS CHKSW3
1132 7410 SKP
1133 5773' JMP CHEXN
1134 1060 CHEX06, TAD STK0
1135 7640 SZA CLA
1136 5344 JMP CHEX07
1137 3104 DCA STKTST
1140 4772' JMS SAME
1141 7410 SKP
1142 4771' JMS RELO
1143 4775' JMS CHEXN1
```

/APT/ IF UNDER APT CONTROL NEXT LOC WILL = JMP APTJ52,

APTJ02,

/APT/

```
1144 4440 CHEX07, LAS
1145 0047 AND SW5 /HALT AFTER TEST
1146 7650 BNA CLA /
1147 5353 JMP ,+4 /NO DO NOT HALT AFTER TEST
```

```

1150 4441      CHKCON
1151 7402      HL7
1152 4770#     JMS PSEUDO      /CHECK TO SEE IF CONSOLE ACTIVE
1153 4440      LAS              /CONSOLE INACTIVE=HALT AFTER TEST
1154 8046      AND SW4          /GO ASK SR QUESTION IF CONSOLE ACTIVE
1155 7640      SZA CLA          /CHANGE STACK LIMITS?
1156 5767#     JMP CHEXA        /YES

      APTJ52,                  /APT/

1157 4774#     JMS CHKSWS      /NO
1160 5206      JMP CHEXDA      /RELOCATE THE PROGRAM
1161 5773#     JMP CHEXN       /INHIBIT PROGRAM RELOCATION

1167 0214
1170 4247
1171 1655
1172 1640
1173 0532
1174 1647
1175 0600
1176 3054
1177 2706
      1200
      PAGE

```

```

/
/RUN THE SELECTED TEST(S) ON THE SELECTED FIELD (STKTST)
/
1200 0000      TEST, 0
/APT/ IF UNDER APT CONTROL NEXT LOC WILL = JMS I IAPTOK.

1201 4404      APTOK0, JMS I KSFCHK /THIS LOC OVERLAID IF APT
1202 7000      NOP /

1203 7300      CLA CLL
1204 1104      TAD STKTST      /UPDATE CDF TEST DATA FIELDS
1205 1137      TAD K6201
1206 3275      DCA TDF1
1207 1275      TAD TDF1
1210 3777#     DCA TDF2
1211 1777#     TAD TDF2
1212 3776#     DCA TDF3
1213 1776#     TAD TDF3
1214 3775#     DCA TDF4
1215 3112      DCA COUNT      /CLEAR ERROR COUNT
1216 1076      TAD RUNTST
1217 7010      RAR
1220 7630      SZL CLA
1221 4273      JMS TEST1      /EXECUTE TEST 1
1222 1076      TAD RUNTST
1223 7012      RTR
1224 7630      SZL CLA
1225 4774#     JMS TEST2      /EXECUTE TEST 2

```

```

1226 1076      TAD RUNTST
1227 7000      RTL
1230 7630      SZL CLA
1231 4773#     JMS TEST3      /EXECUTE TEST 3
1232 1076      TAD RUNTST
1233 7004      RAL
1234 7630      SZL CLA
1235 4772#     JMS TEST4      /EXECUTE TEST 4

/APT/ IF UNDER APT CONTROL NEXT LOC WILL = JMP APTJ53.
APTJ03,        /APT/

1236 4440      LAS
1237 0046      AND SW4        /CHANGE STACK LIMITS?
1240 7640      SZA CLA
1241 5771#     JMP CHEXA      /YES
1242 2262      ISZ FIVE

      APTJ53,                  /APT/

1243 5600      JMP I TEST      /NOT 5 MINUTES YET
1244 1146      TAD MINS
1245 3262      DCA FIVE        /RESTORE TIMER
1246 4441      CHKCON        /CHECK TO SEE IF ACTIVE CONSOLE
1247 5255      JMP .+6         /INACTIVE CONSOLE=DO NOT PRINT PASSES
1250 4426      PRNTH3        /PRINT END OF PASS MESSAGE
1251 1263      PASMES        /POINTER TO MESSAGE
1252 2261      ISZ PASCNT      /ADD 1 TO THE PASS COUNTER
1253 1261      TAD PASCNT      /GET THE COUNTER
1254 4434      PRNT4          /PRINT THE 4 OCTAL DIGITS
1255 4540      JMS I XMESAG
1256 4543      4543
1257 6500      6500
1260 5600      JMP I TEST

1261 0000      PASCNT, 0
1262 0000      FIVE, 0

1263 4304      PASMES, TEXT    "#DHKMC=B PASS #"
1264 1013
1265 1503
1266 5502
1267 4020
1270 0123
1271 2340
1272 0000

/TEST 1
/
/WRITE THE VALUE OF EACH LOCATION INTO ITSELF AND CHECK
1273 0000      TEST1, 0

```

```

1274 3077      DCA    TESTAD      /CLEAR TEST ADDRESS COUNTER
1275 6201      TDF1,  CDF0      /CHANGE TO TEST DATA FIELD
1276 1077      TEST1A, TAD    TESTAD
1277 3477      DCA I   TESTAD      /WRITE MEMORY
1300 2077      ISZ    TESTAD
1301 5276      JMP    TEST1A      /4096 TIMES
1302 1077      TEST1B, TAD    TESTAD      /READ AND CHECK
1303 7041      CIA
1304 1477      TAD I   TESTAD
1305 7640      SZA CLA

/APT/ IF UNDER APT CONTROL NEXT LOC WILL = JMS I IAPTER.

APTE01,      /APT/

1306 5316      JMP    ADDR1      /ADDRESS ERROR
1307 2077      ADDR1,  ISZ    TESTAD
1310 5302      JMP    TEST1A      /CONTINUE READ AND CHECK
1311 1103      TAD    STKPIN
1312 1137      TAD    K6201
1313 3314      DCA    .+1
1314 6201      CDF0
1315 5673      JMP I   TEST1      /CHANGE TO PROGRAM DATA FIELD
                                   /DONE

1316 1077      ADDR1,  TAD    TESTAD
1317 3106      DCA    GOATA      /GOOD
1320 1477      TAD I   TESTAD
1321 3105      DCA    BDATA      /BAD
1322 1103      TAD    STKPIN
1323 1137      TAD    K6201
1324 3325      DCA    .+1
1325 6201      CDF0
1326 4770      JMS    ERR1      /CHANGE TO PROGRAM DATA FIELD
                                   /ADDRESS ERROR TEST1
1327 1104      TAD    STKTST
1330 1137      TAD    K6201
1331 3332      DCA    .+1
1332 6201      CDF0
1333 5307      JMP    ADDR1      /CHANGE TO TEST DATA FIELD

1370 2272
1371 0214
1372 1514
1373 1443
1374 1400
1375 1516
1376 1445
1377 1402
1400 1400      PAGE

```

```

/TEST 2
/
/WRITE THE COMPLEMENT VALUE OF EACH LOCATION INTO ITSELF AND CHECK
/
1400 0000      TEST2, 0

```

```

1401 3077      DCA    TESTAD      /CLEAR TEST ADDRESS COUNTER
1402 6201      TDF2,  CDF0      /CHANGE TO TEST DATA FIELD
1403 1077      TEST2A, TAD    TESTAD
1404 7040      CMA
1405 3477      DCA I   TESTAD      /WRITE MEMORY
1406 2077      ISZ    TESTAD
1407 5203      JMP    TEST2A      /4096 TIMES
1410 1077      TEST2B, TAD    TESTAD      /READ AND CHECK
1411 7001      IAC
1412 1477      TAD I   TESTAD
1413 7640      SZA CLA

/APT/ IF UNDER APT CONTROL NEXT LOC WILL = JMS I IAPTER.

APTE02,      /APT/

1414 5224      JMP    ADDR2      /ADDRESS ERROR
1415 2077      ADDR2,  ISZ    TESTAD
1416 5210      JMP    TEST2B      /CONTINUE READ AND CHECK
1417 1103      TAD    STKPIN
1420 1137      TAD    K6201
1421 3222      DCA    .+1
1422 6201      CDF0
1423 5600      JMP I   TEST2      /CHANGE TO PROGRAM DATA FIELD
                                   /DONE

1424 1077      ADDR2,  TAD    TESTAD
1425 7040      CMA
1426 3106      DCA    GOATA      /GOOD
1427 1477      TAD I   TESTAD
1430 3105      DCA    BDATA      /BAD
1431 1103      TAD    STKPIN
1432 1137      TAD    K6201
1433 3234      DCA    .+1
1434 6201      CDF0
1435 4777      JMS    ERR2      /CHANGE TO PROGRAM DATA FIELD
                                   /ADDRESS ERROR TEST 2
1436 1104      TAD    STKTST
1437 1137      TAD    K6201
1440 3241      DCA    .+1
1441 6201      CDF0
1442 5215      JMP    ADDR2      /CHANGE TO TEST DATA FIELD

```

```

/TEST 3
/
/WRITE THE VALUE OF EACH LOCATION INTO ITSELF AND CHECK BACKWARDS
/
1443 0000      TEST3, 0
1444 3077      DCA    TESTAD      /CLEAR TEST ADDRESS COUNTER
1445 6201      TDF3,  CDF0      /CHANGE TO TEST DATA FIELD
1446 1077      TEST3A, TAD    TESTAD
1447 1114      TAD    M1
1450 3077      DCA    TESTAD
1451 1077      TAD    TESTAD
1452 3477      DCA I   TESTAD      /WRITE MEMORY
1453 1077      TAD    TESTAD
1454 7640      SZA CLA

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```

1455 5246      JMP      TEST3A      /4096 TIMES
1456 1077 TEST3B, TAD      TESTAD
1457 1114      TAD      M1
1460 3077      DCA      TESTAD
1461 1077      TAD      TESTAD      /READ AND CHECK
1462 7041      CIA
1463 1477      TAD I TESTAD
1464 7640      SZA CLA

/APT/ IF UNDER AP CONTROL NEXT LOC WILL = JMS I IAPTER.
APTE03,      /APT/
1465 5276      JMP      ADDR3      /ADDRESS ERROR
1466 1077 ADDR3, TAD      TESTAD
1467 7640      SZA CLA
1470 5256      JMP      TEST3B      /CONTINUE READ AND CHECK
1471 1103      TAD      STKPIN
1472 1137      TAD      K6201
1473 3274      DCA      .+1
1474 6201      CDF0
1475 5643      JMP I TEST3      /CHANGE TO PROGRAM DATA FIELD
                                   /DONE
1476 1077 ADDR3, TAD      TESTAD
1477 3106      DCA      GDATA
1500 1477      TAD I TESTAD      /GOOD
1501 3105      DCA      BDATA
1502 1103      TAD      STKPIN
1503 1137      TAD      K6201
1504 3305      DCA      .+1
1505 6201      CDF0
1506 4776      JMS      ERR3      /CHANGE TO PROGRAM DATA FIELD
1507 1104      TAD      STKTST      /ADDRESS ERROR TEST 3
1510 1137      TAD      K6201
1511 3312      DCA      .+1
1512 6201      CDF0
1513 5266      JMP      ADDR3      /CHANGE TO TEST DATA FIELD

/TEST 4
/
/WRITE THE COMPLEMENT VALUE OF EACH LOCATION INTO ITSELF
/AND CHECK BACKWARDS
/
TEST4, 0
1514 0000      DCA      TESTAD      /CLEAR TEST ADDRESS COUNTER
1515 3077      TDF4, CDF0      /CHANGE TO TEST DATA FIELD
TEST4A, TAD      TESTAD
1516 6201      TAD      M1
1517 1077      DCA      TESTAD
1520 1114      TAD      TESTAD
1521 3077      DCA      TESTAD
1522 1077      TAD      TESTAD
1523 7040      CMA
1524 3077      DCA I TESTAD      /WRITE MEMORY
1525 1077      TAD      TESTAD
1526 7640      SZA CLA
1527 5317      JMP      TEST4A      /4096 TIMES
1530 1077 TEST4B, TAD      TESTAD

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```

1531 1114      TAD      M1
1532 3077      DCA      TESTAD
1533 1077      TAD      TESTAD      /READ AND CHECK
1534 7001      IAC
1535 1477      TAD I TESTAD
1536 7640      SZA CLA

/APT/ IF UNDER APT CONTROL NEXT LOC WILL = JMS I IAPTER.
APTE04,      /APT/
1537 5350      JMP      ADDR4      /ADDRESS ERROR
1540 1077 ADDR4, TAD      TESTAD
1541 7640      SZA CLA
1542 5330      JMP      TEST4B      /CONTINUE READ AND CHECK
1543 1103      TAD      STKPIN
1544 1137      TAD      K6201
1545 3346      DCA      .+1
1546 6201      CDF0
1547 5714      JMP I TEST4      /CHANGE TO PROGRAM DATA FIELD
                                   /DONE
1550 1077 ADDR4, TAD      TESTAD
1551 7040      CMA
1552 3106      DCA      GDATA
1553 1477      TAD I TESTAD      /GOOD
1554 3105      DCA      BDATA
1555 1103      TAD      STKPIN
1556 1137      TAD      K6201
1557 3360      DCA      .+1
1560 6201      CDF0
1561 4775      JMS      ERR4      /CHANGE TO PROGRAM DATA FIELD
1562 1104      TAD      STKTST      /ADDRESS ERROR TEST 4
1563 1137      TAD      K6201
1564 3365      DCA      .+1
1565 6201      CDF0
1566 5340      JMP      ADDR4      /CHANGE TO TEST DATA FIELD

1575 2457
1576 2441
1577 2323
1600      PAGE

/
/SETUP 5 MINUTE TIMER & TEST SELECTED TO RUN
/
1600 7200 RUN0, CLA      /RUN ALL TESTS
1601 1231      TAD      MINS0
1602 3146      DCA      MINS
1603 1236      TAD      K6003
1604 5777      JMP      EXTAD0
1605 7200 RUN1, CLA      /RUN ONLY TEST 1
1606 1232      TAD      MINS1
1607 3146      DCA      MINS

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1610 7001      IAC
1611 5777'     JMP      EXTAD0
1612 7300      RUN2,  CLA CLL      /RUN ONLY TEST 2
1613 1233      TAD      MINS2
1614 3146      DCA      MINS
1615 7009      IAC RAL
1616 5777'     JMP      EXTAD0
1617 7200      RUN3,  CLA      /RUN ONLY TEST 3
1620 1234      TAD      MINS3
1621 3146      DCA      MINS
1622 7132      STL RTR
1623 5777'     JMP      EXTAD0
1624 7200      RUN4,  CLA      /RUN ONLY TEST 4
1625 1235      TAD      MINS4
1626 3146      DCA      MINS
1627 7130      STL RAR
1630 5777'     JMP      EXTAD0

1631 7100      MINS0, -700
1632 3500      MINS1, -4300
1633 3500      MINS2, -4300
1634 6000      MINS3, -2000
1635 6000      MINS4, -2000
1636 6003      K6003, 6003
1637 6203      K6203, 6203

/
/RETURN IF PROGRAM IS IN SELECTED STACK
/RETURN+1 IF PROGRAM IS NOT IN SELECTED STACK
/
1640 0000      SAME,  0
1641 1103      TAD      STKPIN
1642 7041      CIA
1643 1104      TAD      STKTST
1644 7640      SZA CLA
1645 2240      ISZ      SAME      /PROG NOT IN SEL STACK
1646 5640      JMP I     SAME

/
/CHECK PROGRAM RELOCATION SWITCH
/RETURN IF RELOCATE, RETURN+1 IF INHIBIT RELOCATION
/
1647 0000      CHKSW3, 0
1650 4440      LAS
1651 0045      AND      SW3
1652 7640      SZA CLA
1653 2247      ISZ      CHKSW3      /INHIBIT RELOCATION
1654 5647      JMP I     CHKSW3

/
/RELOCATE THE PROGRAM
/
1655 0000      REL0,  0
1656 7200      CLA
1657 3112      DCA      COUNT      /CLEAR ERROR COUNTER
1660 3107      DCA      MOVE
1661 1137      TAD      K6201

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1662 1103      TAD      STKPIN
1663 3274      DCA      RELO2
1664 1137      TAD      K6201
1665 1104      TAD      STKTST
1666 3276      DCA      RELO3
1667 1274      TAD      RELO2
1670 3301      DCA      RELO4
1671 1237      TAD      K6203
1672 1104      TAD      STKTST
1673 3312      DCA      RELO5
1674 6201      RELO2,  CDF0      /MOVE FROM DATA FIELD
1675 1507      TAD I     MOVE
1676 6201      RELO3,  CDF0      /MOVE TO DATA FIELD
1677 3507      DCA I     MOVE
1700 1507      TAD I     MOVE
1701 6201      RELO4,  CDF0      /MOVE FROM DATA FIELD
1702 7041      CIA
1703 1507      TAD I     MOVE
1704 7640      SZA CLA

/APT/ IF UNDER APT CONTROL NEXT LOC WILL = JMS I IAPTR.
APTE00,      /APT/

1705 4776'     JMS      ERRM      /MOVE ERROR
1706 2107      ISZ      MOVE
1707 5274      JMP      RELO2
1710 1112      TAD      COUNT
1711 7650      SNA CLA      /SKIP IF MOVE ERROR
1712 6203      RELO5,  CDF0      /CHANGE TO NEW PROG FIELD
1713 5655      JMP I     RELO

/
/INCREMENT CONTROL OF UPPER STACKS NOT TESTED AND/OR
/STACKS NOT IN THE SYSTEM
/
1714 0000      0      /RETURN ADDRESS
1715 2051      ABOVE,  ISZ      STACK1
1716 2052      ISZ      STACK2
1717 2053      ISZ      STACK3
1720 2054      ISZ      STACK4
1721 2055      ISZ      STACK5
1722 2056      ISZ      STACK6
1723 2057      ISZ      STACK7
1724 5714      JMP I     ABOVE+1

/
/INCREMENT CONTROL OF LOWER STACKS NOT TESTED
/
1725 2056      ISZ      STACK6
1726 2055      ISZ      STACK5
1727 2054      ISZ      STACK4
1730 2053      ISZ      STACK3
1731 2052      ISZ      STACK2
1732 2051      ISZ      STACK1
1733 2050      ISZ      STACK0

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```

1734 5735      BELOW, JMP I  .+1
1735 0000      B
/
/CLEAR ALL STACKS OR STACKS TO BE TESTED
/
1736 3057      DCA    STACK7
1737 3056      DCA    STACK6
1740 3055      DCA    STACK5
1741 3054      DCA    STACK4
1742 3053      DCA    STACK3
1743 3052      DCA    STACK2
1744 3051      DCA    STACK1
1745 3050      DOWN,  DCA    STACK0
1746 5747      JMP I  .+1
1747 0000      B
1750 5336      JMP    .+12      /RETURN ADDRESS
/
/CLEAR ALL STACK SELECTION CONTROLS
/
/OBTAIN -SSL (MINUS STARTING STACK LIMIT)
/
1751 0000      MSSL,  B
1752 1101      TAD    SSL
1753 7112      CLL RTR
1754 7010      RAR
1755 7041      CIA
1756 5751      JMP I  MSSL
/
1776 2475
1777 0205
2000      PAGE

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```

/
/CONVERT OCTAL NUMBERS FOR TYPEOUT
/
2000 0000      SIXTY, 0
2001 7300      CLA CLL
2002 1600      TAD I  SIXTY      /GET ADDRESS OF OPERAND
2003 3237      DCA    SIXTY0
2004 2200      ISZ    SIXTY
2005 1600      TAD I  SIXTY      /GET STORAGE ADDRESS
2006 3240      DCA    SIXTY1
2007 2200      ISZ    SIXTY      /CORRECT RETURN ADDRESS
2010 1131      TAD    K77
2011 7040      CMA
2012 0637      AND I  SIXTY0      /AC=7700
2013 7112      CLL RTR      /AND OPERAND FIRST 2 DIGITS
2014 7012      RTR
2015 7012      RTR
2016 4224      JMS    CNV      /POSITION FIRST 2 DIGITS
2017 2240      ISZ    SIXTY1      /CONVERT DIGITS FOR TYPEOUT
2020 1131      TAD    K77      /INCREMENT STORAGE ADDRESS
2021 0637      AND I  SIXTY0      /AND OPERAND SECOND 2 DIGITS
2022 4224      JMS    CNV      /CONVERT DIGITS FOR TYPEOUT
2023 5600      JMP I  SIXTY
2024 0000      CNV,  B
2025 3241      DCA    SIXTY2      /SAVE DIGITS

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```

2026 1241      TAD    SIXTY2
2027 7106      CLL RTL
2030 7000      RAL
2031 0325      AND    K707      /AND LEFT DIGIT
2032 1241      TAD    SIXTY2
2033 0325      AND    K707      /AND RIGHT DIGIT
2034 1326      TAD    K6060
2035 3640      DCA I  SIXTY1      /STORE CONVERTED DIGITS
2036 5624      JMP I  CNV
/
/
/TELETYPE OUTPUT ROUTINE WITH BELL
/
2042 0000      MESSAGE, 0
/
2043 7200      CLA
2044 1022      TAD    HCW2      /APT/
2045 7700      SMA CLA      /APT/UNDER APT CONTROL?
2046 5252      JMP      APT000    /APT/SKP IF YES,
2047 1242      TAD    MESSAGE    /APT/
2050 3777'     DCA    APTER      /APT/FORCE AN ERROR CALL TO APT.
2051 5776'     JMP      APTER+1    /APT/
/
APT000,      /APT/
/
2052 7240      STA
2053 1242      TAD    MESSAGE    /FIRST WORD -1
2054 3010      DCA    10
/
APT001,
/
2055 1410      TAD I  10
2056 3267      DCA    MSGRGT
2057 1267      TAD    MSGRGT
2060 7112      CLL RTR
2061 7012      RTR
2062 7012      RTR
2063 4270      JMS    TYPECH      /POSITION FIRST CHARACTER
2064 1267      TAD    MSGRGT      /TYPEOUT FIRST CHARACTER
2065 4270      JMS    TYPECH      /TYPEOUT SECOND CHARACTER
/
/*APT*/      JMP    MESSAGE+4    /CONTINUE TYPING
/
2066 5255      JMP      APT001    /APT/CONTINUE TYPING.
/
2067 0000      MSGRGT, 0
2070 0000      TYPECH, 0
2071 0131      AND    K77
2072 7450      BNA
2073 5410      JMP I  10      /IS IT END OF MESSAGE?
/RETURN TO PROGRAM

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```

2074 1120 TAD M34 /SUBTRACT 34
2075 7440 SZA
2076 5301 JMP ,+3
2077 1132 TAD K207 /CODE IS BELL
2100 5321 JMP MTP
2101 1117 TAD M4 /SUBTRACT 4
2102 7500 SNA /CODE LESS THAN 40?
2103 5306 JMP ,+3 /NO
2104 1324 TAD K340 /YES, ADD 300, CODE IS ALPHA
2105 5321 JMP MTP
2106 1116 TAD M3 /SUBTRACT 3
2107 7440 SZA
2110 5313 JMP ,+3
2111 1133 TAD K212 /CODE IS LINE FEED
2112 5321 JMP MTP
2113 1115 TAD M2 /SUBTRACT 2
2114 7440 SZA
2115 5320 JMP ,+3
2116 1134 TAD K215 /CODE IS CR
2117 7410 SKP
2120 1323 TAD K245 /ADD 200 TO OTHER CODES >40
2121 4436 MTP, TYPE /TYPEOUT CHARACTER IN AC
2122 5670 JMP I TYPECH

2123 0245 K245, 245
2124 0340 K340, 340
2125 0707 K707, 707
2126 6060 K6060, 6060

```

```

/TYPEOUT "PROGRAM IS IN SELECTED FIELD"
/
2127 1024 PINF, TAD NOTTY /GET THE TERMINAL FLAG
2130 7640 SZA CLA /IS THERE A TERMINAL ON THE SYSTEM
2131 5775' JMP CHEXA /NO-GO SETUP THE SR AGAIN
2132 4540 JMS I XMSAG
2133 4543 TEXT "%PROGRAM IN SELECTED FIELD"
2134 2022
2135 1707
2136 2201
2137 1540
2140 1116
2141 4023
2142 0514
2143 0503
2144 2405
2145 0440
2146 0611
2147 0514
2150 0400
2151 5775' JMP CHEXA /SETUP SWITCHES AGAIN

```

/TYPEOUT "NONE" FOR NO LEGAL STACK SELECTION

```

2152 1024 NOSTK, TAD NOTTY /GET THE TERMINAL FLAG
2153 7640 SZA CLA /IS THERE A TERMINAL ON THE SYSTEM
2154 5775' JMP CHEXA /NO-ABORT MESSAGE AND GO SETUP SR AGAIN
2155 4540 JMS I XMSAG
2156 1617 TEXT "NONE"
2157 1605
2160 0000
2161 5775' JMP CHEXA /SETUP SWITCHES AGAIN

2175 0214
2176 4027
2177 4026
2200 PAGE

```

```

/ERROR ROUTINE (BELL ON ERROR HAS PRIORITY)
/
2200 0000 RETURN, 0 /PROGRAM RETURN ADDRESS
2201 6002 CODERR, IOF
2202 4440 LAS /CHECK FOR BELL ON ERROR
2203 0044 AND SW2
2204 7650 SNA CLA
2205 5215 JMP NOBELL
2206 1024 RBELL, TAD NOTTY /GET TERMINAL FLAG
2207 7640 SZA CLA /IS THERE A TERMINAL ON THE SYSTEM
2210 5213 JMP ,+3 /NO-DO NOT RING BELL BUT RETURN TO PRGRAM
2211 1132 TAD K207 /BELL CODE
2212 4436 TYPE /RING BELL
2213 4404 JMS I KSFCMK /CHECK FOR CONSOLE RECEIVE FLAG
2214 5600 JMP I RETURN
2215 4440 NOBELL, LAS /CHECK FOR INHIBIT TYPEOUT
2216 0043 AND SW1
2217 7640 SZA CLA
2220 5251 JMP STOP /INHIBIT TYPEOUT
2221 6224 RIF /READ INST FIELD
2222 7012 RTR
2223 7010 RAR
2224 0121 AND K7
2225 1136 TAD K4060
2226 3242 OCA ERROR0
2227 1200 TAD RETURN
2230 1114 TAD M1
2231 3113 OCA ERRLOC
2232 4541 JMS I XSIXTY
2233 0113 ERRLOC
2234 2243 ERROR1
2235 1024 TAD NOTTY /GET THE TERMINAL FLAG
2236 7640 SZA CLA /IS THERE A TERMINAL ON SYSTEM
2237 5650 JMP I ADDER /NO-GO TO APPROPRIATE ERROR ROUTINE
2240 4540 JMS I XMSAG /TYPEOUT ERROR LOCATION
2241 4543
2242 0000 ERROR0, 0 /FIELD
2243 0000 ERROR1, 0
2244 0000 /PROGRAM LOCATION OF ERROR JMS

```

```

2243 4040      4040
2244 0000      0000
2247 5630      JMP I  .+1      /TYPEOUT ERROR
2250 0000      ADDER, 0      /ADDRESS OF ERROR TYPEOUT

2251 4440      STOP, LAB      /HALT AFTER ERROR
2252 0042      AND SW0
2253 7650      SNA CLA
2254 5163      JMP LIMIT      /INHIBIT ERROR HALT
2255 4441      CHKCON      /CHECK TO SEE IF CONSOLE ACTIVE
2256 7610      SKP CLA      /CONSOLE INACTIVE=HALT WITH AC=ERR LOC
2257 5777      JMP CONHLT      /CONSOLE ACTIVE=GO TYPE ERR LOCATION
2260 7240      CLA CMA      /SUBTRACT 1 FROM JMS ERROR
2261 1800      TAD RETURN      /GET THE ERR ADDRESS +1
2262 7402      HLT      /HALT AFTER ERROR=AC=ERROR LOCATION
2263 7200      LIMIT, CLA
2264 4404      JMS I KSFCMK      /CHECK FOR CONSOLE RECEIVE FLAG
2265 4440      LAB
2266 0046      AND SW4      /CHANGE STACK LIMITS?
2267 7640      SZA CLA
2270 5776      JMP CHEXA      /YES
2271 5600      JMP I RETURN      /NO

```

```

/
/ADDRESS ERROR TEST 1
/
2272 0000      ERR1, 0
2273 2112      ISZ COUNT      /ADDRESS ERROR OCCURRED
2274 7410      SKP
2275 5273      JMP  .-2
2276 7200      CLA
2277 1272      TAD ERR1
2300 3543      DCA I XRETUR      /STORE RETURN ADDRESS
2301 1304      TAD  .+3
2302 3545      DCA I XADDER      /STORE ERROR TYPEOUT ADDRESS
2303 7410      SKP
2304 3400      PERR1
2305 1375      TAD (6100
2306 3774      DCA Z24      /TEST 1
2307 4440      ERR1A, LAB
2310 0044      AND SW2      /BELL ON ERROR?
2311 7640      SZA CLA
2312 5206      JMP RBELL      /YES
2313 4440      LAB
2314 0043      AND SW1      /INHIBIT ERROR TYPEOUT?
2315 7640      SZA CLA
2316 5544      JMP I XSTOP      /YES
2317 2073      ISZ HEAD1
2320 7410      SKP
2321 4773      JMS HEAD12      /TYPEOUT ERROR HEADING
2322 5542      JMP I XCODER      /GO TO ERROR ROUTINE
/
/ADDRESS ERROR TEST 2

```

```

/
2323 0000      ERR2, 0
2324 2112      ISZ COUNT      /ADDRESS ERROR OCCURRED
2325 7410      SKP
2326 5324      JMP  .-2
2327 7200      CLA
2330 1323      TAD ERR2
2331 3543      DCA I XRETUR      /STORE RETURN ADDRESS
2332 1335      TAD  .+3
2333 3545      DCA I XADDER      /STORE ERROR TYPEOUT ADDRESS
2334 7410      SKP
2335 3400      PERR1
2336 1372      TAD (6200
2337 3774      DCA Z24      /TEST 2
2340 5307      JMP ERR1A

```

```

2372 6200
2373 2510
2374 3434
2375 6100
2376 0214
2377 2400
2400

```

PAGE

/HALT AFTER ERROR MESSAGE-REPORTED IF SR0=1

```

2400 4426      CONHLT, PRNTHS      /GO REPORT ERROR MESSAGE
2401 2407      ERRMES      /POINTER TO ERROR MESSAGE
2402 7240      CLA CMA      /PICK UP ERROR LOCATION -1
2403 1777      TAD RETURN      /
2404 4434      PRNT4      /PRINT THE 4 OCTAL DIGITS
2405 4776      JMS PSEUDO      /FORCE PROGRAM TO SR QUESTION
2406 5775      JMP LIMIT      /GO CHECK SR 4 FOR STACK CHANGES

2407 4323      ERRMES, TEXT      "SR0=1-HALT AFTER ERROR=ERROR DETECTED AT LOCATION "
2410 2760
2411 7561
2412 5510
2413 0114
2414 2440
2415 0106
2416 2405
2417 2240
2420 0522
2421 2217
2422 2255
2423 0522
2424 2217
2425 2240
2426 0405
2427 2405
2430 0324
2431 0504
2432 4001

```

2433 2440
2434 1417
2435 0301
2436 2411
2437 1716
2440 4000

```

/
/ADDRESS ERROR TEST 3
/
ERR3, 0
      ISZ   COUNT      /ADDRESS ERROR OCCURRED
      SKP
      JMP   ,-2
      CLA
      TAD   ERR3
      DCA I  XRETUR     /STORE RETURN ADDRESS
      TAD   ,+3
      DCA I  XADDR      /STORE ERROR TYPEOUT ADDRESS
      SKP
      PERR1
      TAD   (6300
      DCA   Z24
      JMP   ERR1A      /TEST 3
2441 0000
2442 2112
2443 7410
2444 5242
2445 7200
2446 1241
2447 3543
2450 1253
2451 3545
2452 7410
2453 3400
2454 1374
2455 3773
2456 5772

```

```

/
/ADDRESS ERROR TEST 4
/
ERR4, 0
      ISZ   COUNT      /ADDRESS ERROR OCCURRED
      SKP
      JMP   ,-2
      CLA
      TAD   ERR4
      DCA I  XRETUR     /STORE RETURN ADDRESS
      TAD   ,+3
      DCA I  XADDR      /STORE ERROR TYPEOUT ADDRESS
      SKP
      PERR1
      TAD   (6400
      DCA   Z24
      JMP   ERR1A      /TEST 4
2457 0000
2460 2112
2461 7410
2462 5242
2463 7200
2464 1257
2465 3543
2466 1271
2467 3545
2470 7410
2471 3400
2472 1371
2473 3773
2474 5772

```

```

/
/RELOCATION MOVE ERROR OCCURRED
/
ERRM, 0
      ISZ   COUNT      /RELO ERROR OCCURRED
      SKP
      JMP   ,-2
      CLA
      TAD   ERRM
      DCA I  XRETUR     /STORE RETURN ADDRESS
      TAD   ,+3
      DCA I  XADDR      /STORE ERROR TYPEOUT ADDRESS
2475 0000
2476 2112
2477 7410
2500 5276
2501 7200
2502 1275
2503 3543
2504 1307
2505 3545

```

```

2506 5542      JMP I  XCODER
2507 3436      PERRM

/
/TYPEOUT TEST 1 OR 2 ERROR HEADING
/
HEAD12, 0
      TAD   NOTTY      /GET THE TERMINAL FLAG
      SZA   CLA        /IS THERE A TERMINAL ON THE SYSTEM
      JMP I  HEAD12     /NO-RETURN WITHOUT TELETYPE OUTPUT
      JMS I  XMSAG
      TEXT  "X*PR LOC  ADDR  GOOD BAD TEST"
2510 0000
2511 1024
2512 7640
2513 5710
2514 4540
2515 4543
2516 2022
2517 4014
2520 1703
2521 4040
2522 4001
2523 0404
2524 2240
2525 4040
2526 0717
2527 1704
2530 4040
2531 0201
2532 0440
2533 4024
2534 0523
2535 2400
2536 5710      JMP I  HEAD12

2571 6400
2572 2307
2573 3434
2574 6300
2575 2263
2576 4247
2577 2200
2600      PAGE

/
/TYPEOUT PROGRAM TITLE
/
TITLE, 0
      TAD   NOTTY      /CHECK TO SEE IF CONSOLE AVAILABLE
      SZA   CLA        /IS THERE A CONSOLE TERMINAL?
      JMP I  TITLE     /NO-DO NOT TRY TO TYPE MESSAGE
      JMS I  XMSAG
      TEXT  "X*EAS-E EXT MEM ADDR TEST#"
2600 0000
2601 1024
2602 7640
2603 5600
2604 4540
2605 4543
2606 4305
2607 0170
2610 5505
2611 4005
2612 3024
2613 4015
2614 0515

```

2615 4001
2616 0404
2617 2240
2620 2405
2621 2324
2622 4300
2623 5600

JMP I TITLE

/
/TYPEOUT TO SET SWITCHES

2624 0000 SETSW, 0
2625 1024 TAD NOTTY /GET THE CONSOLE FLAG
2626 7640 SZA CLA /IS THERE A CONSOLE TERMINAL
2627 5244 JMP SETSW1 /NO-HALT TO ALLOW SR SETTINGS
2630 4540 JMS I XMSAG
2631 4543 TEXT "%SETUP SR & CONTX"
2632 2305
2633 2425
2634 2040
2635 2322
2636 4046
2637 4003
2640 1716
2641 2445
2642 4300
2643 4441 CHKCON /CHECK TO SEE IF CONSOLE ACTIVE
2644 7402 SETSW1, HLT /CONSOLE INACTIVE-SET SR AND CONT
2645 4777 JMS PSEUDO /ASK SR QUESTION IF ON ACTIVE CONSOLE
2646 5624 JMP I SETSW

/
/TYPEOUT 'NO PROGRAM RELOCATION WILL OCCUR'

2647 0000 PNOREL, 0
2650 1024 TAD NOTTY /GET THE CONSOLE FLAG
2651 7640 SZA CLA /IS THERE A TERMINAL
2652 5303 JMP Z0+1 /NO ABORT ROUTINE
2653 4540 JMS I XMSAG
2654 4543 TEXT "%NO RELOCATION, PROG IN STACK "
2655 1617
2656 4022
2657 0514
2660 1703
2661 0124
2662 1117
2663 1654
2664 4020
2665 2217
2666 0740
2667 1116
2670 4023
2671 2401
2672 0313
2673 4000
2674 6224 RIF

2675 7106 CLL RTL
2676 7004 RAL
2677 1376 TAD (6000
2700 3302 DCA Z0
2701 4540 JMS I XMSAG
2702 0000 Z0, 0
2703 7240 STA
2704 3073 DCA HEAD1 /RESET ERROR HEADING
2705 5647 JMP I PNOREL

/PROGRAM RELOCATION WILL OCCUR

2706 0000 PREL, 0
2707 1024 TAD NOTTY /GET THE CONSOLE TERMINAL FLAG
2710 7640 SZA CLA /IS THERE A TERMINAL ON SYSTEM
2711 5326 JMP PREL1 /NO-ABORT TYPEOUT THEN RETURN
2712 4540 JMS I XMSAG
2713 4543 TEXT "%PROG WILL RELOCATE"
2714 2022
2715 1707
2716 4027
2717 1114
2720 1440
2721 2205
2722 1417
2723 0301
2724 2405
2725 0000

2726 7240 PREL1, STA
2727 3073 DCA HEAD1 /RESET ERROR HEADING
2730 5706 JMP I PREL

/FIND HIGHEST STACK NUMBER IN THIS SYSTEM

2731 0000 HIGHEST, 0
2732 7300 CLA CLL
2733 3100 DCA KBINT /CLEAR HIGH STACK COUNTER
2734 6211 CDF1
2735 4775 JMS CSS /CHECK FOR FIELD 1
2736 6221 CDF2
2737 4775 JMS CSS /CHECK FOR FIELD 2
2740 6231 CDF3
2741 4775 JMS CSS /CHECK FOR FIELD 3
2742 6241 CDF4
2743 4775 JMS CSS /CHECK FOR FIELD 4
2744 6251 CDF5
2745 4775 JMS CSS /CHECK FOR FIELD 5
2746 6261 CDF6
2747 4775 JMS CSS /CHECK FOR FIELD 6
2750 6271 CDF7
2751 4775 JMS CSS /CHECK FOR FIELD 7

```

2752 5731 KHIGH, JMP I HIGHST
2775 3000
2776 6000
2777 4247
3000
PAGE
/
/CHECK IF SELECTED STACK IS IN SYSTEM
/
3000 0000 CSS, 0
3001 7300 CLA CLL
3002 6224 RIF
3003 1137 TAO K6201
3004 3210 DCA CS30
3005 1114 TAO M1
3006 3615 DCA I CHECK
3007 1615 TAO I CHECK
3010 6201 CSSB, CDF 00 /PROGRAM DATA FIELD
3011 7650 SNA CLA /SKIP IF STACK IS IN SYSTEM
3012 5777 JMP KHIGH
3013 2100 ISZ KBINT /INCREMENT STACK COUNTER
3014 5600 JMP I CSS
3015 3016 CHECK, CHECK0
3016 0000 CHECK0, 0
/
/TYPEOUT NUMBER OF STACKS IN SYSTEM
/
3017 0000 TSTSYS, 0
3020 1024 TAD NOTTY /GET TERMINAL FLAG
3021 7640 SZA CLA /IS THERE A TERMINAL ON THE SYSTEM
3022 5617 JMP I TSTSYS /NO-ABORT TELETYPE MESSAGE
3023 4540 JMS I XMESAG
3024 4543
3025 0000
3026 1100 TAD KBINT
3027 7001 IAC
3030 4436 TYPE /TYPEOUT NUMBER
3031 4540 JMS I XMESAG
3032 4023 TEXT " STACKS IN THIS SYSTEM"
3033 2401
3034 0313
3035 2340
3036 1116
3037 4024
3040 1011
3041 2340
3042 2331
3043 2324
3044 0515
3045 0000
3046 5617 JMP I TSTSYS

```

```

/
/TYPEOUT CHARACTER IN THE AC AND A SPACE
/
3047 0000 TYPESP, 0
3050 4436 TYPE /TYPEOUT CHAR IN AC
3051 1135 TAD K240
3052 4436 TYPE /TYPE A SPACE
3053 5647 JMP I TYPESP
/
/RESTORE STACKS FOR RELOCATION
/
3054 0000 RESTK, 0
3055 7200 CLA
3056 1050 TAD STACK0
3057 3060 DCA STK0
3060 1051 TAD STACK1
3061 3061 DCA STK1
3062 1052 TAD STACK2
3063 3062 DCA STK2
3064 1053 TAD STACK3
3065 3063 DCA STK3
3066 1054 TAD STACK4
3067 3064 DCA STK4
3070 1055 TAD STACK5
3071 3065 DCA STK5
3072 1056 TAD STACK6
3073 3066 DCA STK6
3074 1057 TAD STACK7
3075 3067 DCA STK7
3076 5654 JMP I RESTK
/
/TYPEOUT STACKS SELECTED FOR TESTING
/
3077 0000 TOSEL, 0
3100 1024 TAD NOTTY /GET TERMINAL FLAG
3101 7640 SZA CLA /IS THERE A TERMINAL ON SYSTEM
3102 5677 JMP I TOSEL /NO-ABORT TELETYPE MESSAGE
3103 4540 JMS I XMESAG
3104 4543 TEXT "X#STACKS SEL'D ARE "
3105 2324
3106 0103
3107 1323
3110 4023
3111 0514
3112 4704
3113 4001
3114 2205
3115 0000
3116 1057 TAD STACK7
3117 7640 SZA CLA
3120 5323 JMP ,+3
3121 1376 TAD (207

```

3122	4247	JMS	TYPE8P	/STACK 7 IS SELECTED
3123	1056	TAD	STACK6	
3124	7640	8ZA	CLA	
3125	5330	JMP	,+3	
3126	1375	TAD	(266	
3127	4247	JMS	TYPE8P	/STACK 6 IS SELECTED
3130	1055	TAD	STACK5	
3131	7640	8ZA	CLA	
3132	5335	JMP	,+3	
3133	1374	TAD	(265	
3134	4247	JMS	TYPE8P	/STACK 5 IS SELECTED
3135	1054	TAD	STACK4	
3136	7640	8ZA	CLA	
3137	5342	JMP	,+3	
3140	1373	TAD	(264	
3141	4247	JMS	TYPE8P	/STACK 4 IS SELECTED
3142	1053	TAD	STACK3	
3143	7640	8ZA	CLA	
3144	5347	JMP	,+3	
3145	1372	TAD	(263	
3146	4247	JMS	TYPE8P	/STACK 3 IS SELECTED
3147	1052	TAD	STACK2	
3150	7640	8ZA	CLA	
3151	5354	JMP	,+3	
3152	1371	TAD	(262	
3153	4247	JMS	TYPE8P	/STACK 2 IS SELECTED
3154	1051	TAD	STACK1	
3155	7640	8ZA	CLA	
3156	5361	JMP	,+3	
3157	1370	TAD	(261	
3160	4247	JMS	TYPE8P	/STACK 1 IS SELECTED
3161	1050	TAD	STACK0	
3162	7640	8ZA	CLA	
3163	5366	JMP	,+3	
3164	1367	TAD	(260	
3165	4247	JMS	TYPE8P	/STACK 0 IS SELECTED
3166	5677	JMP	I TOSEL	

3167 0260
3170 0261
3171 0262
3172 0263
3173 0264
3174 0265
3175 0266
3176 0267
3177 2752
3200

PAGE

/

/TWO SPECIAL SCOPE LOOPS

/

3200 4440 LOOP1, LAB /SWITCH ADDRESS

3201	3206	DCA	SWAD	
3202	1206	TAD	SWAD	
3203	3606	DCA I	SWAD	
3204	1606	TAD I	SWAD	
3205	5200	JMP	LOOP1	
3206	0000	SWAD,	0	
3207	4440	LOOP2,	LAB	/READ LOWER LIMIT
3210	3233	DCA	FIRST	
3211	7402	HLT		/SET SR FOR UPPER LIMIT
3212	4440	LAB		/READ UPPER LIMIT
3213	3234	DCA	LAST	
3214	1233	LOOP2A, TAD	FIRST	
3215	3235	DCA	SWAD0	
3216	1235	LOOP2B, TAD	SWAD0	
3217	3635	DCA I	SWAD0	
3220	1635	TAD I	SWAD0	
3221	7200	CLA		
3222	1235	TAD	SWAD0	
3223	7041	CIA		
3224	1234	TAD	LAST	
3225	7650	SNA CLA		
3226	5214	JMP	LOOP2A	
3227	2235	ISZ	SWAD0	
3230	5216	JMP	LOOP2B	
3231	7402	HLT		/HALT RESULTED IN ILLEGAL LIMITS
3232	5207	JMP	LOOP2	
3233	0000	FIRST,	0	
3234	0000	LAST,	0	
3235	0000	SWAD0,	0	

3400 PAGE

3400	1104	PERR1,	TAD	STKTST	
3401	7112		CLL	RTR	
3402	7010		RAR		
3403	1136		TAD	K4060	
3404	3222		DCA	Z20	/FIELD OF ERROR
3405	4541		JMS I	X5IXTY	
3406	0077		TESTAD		
3407	3423		Z21		/FAILING ADDRESS
3410	4541		JMS I	X5IXTY	
3411	0106		GDATA		
3412	3426		Z22		/GOOD
3413	4541		JMS I	X5IXTY	
3414	0105		BDATA		
3415	3431		Z23		/BAD
3416	1024		TAD	NOTTY	/GET THE TERMINAL FLAG
3417	7640		8ZA	CLA	/IS THERE A TERMINAL ON SYSTEM
3420	5271		JMP	ADDER	/NO-HALT WITH ERROR INFO IN AC
3421	4540		JMS I	XME0AG	

```

3422 0000    Z20, 0
3423 0000    Z21, 0
3424 0000
3425 4040
3426 0000    Z22, 0
3427 0000
3430 4040
3431 0000    Z23, 0
3432 0000
3433 4040
3434 0000    Z24, 0
3435 5544    JMP I XSTOP

3436 1104    PERRM, TAD STKTST
3437 7112    CLL RTR
3440 7010    RAR
3441 1136    TAD K0060
3442 3262    DCA Z10
3443 4541    JMS I XSIXTY
3444 0107    MOVE
3445 3463    Z11
3446 1024    TAD NOTTY
3447 7640    BZA CLA
3450 5325    JMP RELERR
3451 4540    JMS I XMESAG
3452 2205    TEXT
3453 1417
3454 4005
3455 2222
3456 4001
3457 2440
3460 0000
3461 4540    JMS I XMESAG
3462 0000    Z10, 0
3463 0000    Z11, 0
3464 0000
3465 0000
3466 7240    STA
3467 3073    DCA HEAD1
3470 5544    JMP I XSTOP

```

/ERROR ROUTINE FOR ADDRESS ERRORS ON A SYSTEM WITH NO CONSOLE TERMINAL.
/ERRORS WILL BE REPORTED BY HALTS WITH ERROR INFO IN THE AC. REFER
/TO THE COMMENTS AT EACH HALT FOR THE ERROR INFO SUPPLIED.

```

3471 1777'   ADDERR, TAD ERROR0
3472 0121    AND K7
3473 7402    HLT
3474 7200    CLA
3475 1113    TAD ERRLOC
3476 7402    HLT
3477 7200    CLA
3500 1222    TAD Z20

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```

3501 0121    AND K7
3502 7402    HLT
3503 7200    CLA
3504 1077    TAD TESTAD
3505 7402    HLT
3506 7200    CLA
3507 1106    TAD GOATA
3510 7402    HLT
3511 7200    CLA
3512 1105    TAD BDATA
3513 7402    HLT
3514 7200    CLA
3515 1234    TAD Z24
3516 7112    CLL RTR
3517 7012    RTR
3520 7012    RTR
3521 0121    AND K7
3522 7402    HLT
3523 7200    CLA
3524 5544    JMP I XSTOP

```

/ERROR ROUTINE FOR RELOCATION ERRORS. ERROR INFO WILL BE PROVIDED BY
/ERROR HALTS WITH THE ERROR INFORMATION IN THE AC. REFER TO THE
/COMMENTS AT EACH HALT FOR ERROR INFORMATION SUPPLIED.

```

3525 1777'   RELERR, TAD ERROR0
3526 0121    AND K7
3527 7402    HLT
3530 7200    CLA
3531 1113    TAD ERRLOC
3532 7402    HLT
3533 7200    CLA
3534 1262    TAD Z10
3535 0121    AND K7
3536 7402    HLT
3537 7200    CLA
3540 1107    TAD MOVE
3541 7402    HLT
3542 7200    CLA
3543 5544    JMP I XSTOP

```

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3577 2242    PAGE
3600 6002    APTIZ, IOF
3601 4441    CHKCON
3602 5214    JMP APTIZ1
3603 3024    DCA NOTTY
3604 7240    CLA CMA
3605 3777'   DCA PASCNT
3606 1376    TAD (JMS I K8FCHK

```

```

3607 3775# DCA APTOK0 /C8/
/C8/ THE NEXT LOC WILL = NOP AFTER BEING USED ONCE.

3610 4774# JMS C8SM /C8/GO SAVE PG 37 OF FLD 1.
3611 1373 TAD (7000 /C8/MODIFY ABOVE LOC TO: NOP.
3612 3210 DCA .+2 /C8/
3613 5302 JMP APTIZ0 /C8/

3614 1024 APTIZ1, TAD NOTTY /GET CONSOLE FLAG
3615 7650 SNA CLA /IS THERE A CONSOLE ON THE SYSTEM
3616 5223 JMP .+5 /YES-GO GET THE FIELD LIMITS
3617 1020 TAD PSR /NO-SETUP LIMITS TO 07
3620 0372 AND (7700 /MASK TO SR BITS
3621 1121 TAD KY /ADD FIELD LIMITS
3622 3020 DCA PSR /SAVE SR AS XX07
3623 4771# JMS APTPL /APT/GO GET FIELD LIMITS.
3624 7200 CLA /APT/
3625 1022 TAD MCW2 /APT/RUN UNDER APT CONTROL?
3626 7700 SNA CLA /APT/SKP IF YES.
3627 5302 JMP APTIZ0 /APT/
3630 1370 TAD (JMS I IAPTK /SETUP FOR APT CONTROL
3631 3775# DCA APTOK0 /APT/
3632 1373 TAD (7000 /APT/MODIFY SOME LOCS TO: NOP.
3633 3767# DCA APTN00 /APT/
3634 1373 TAD (7000 /APT/
3635 3766# DCA APTN01 /APT/
3636 1373 TAD (7000 /APT/
3637 3765# DCA APTN02 /APT/
3640 1373 TAD (7000 /APT/
3641 3764# DCA APTN03 /APT/
3642 1373 TAD (7000 /APT/
3643 3763# DCA APTN04 /APT/
3644 1373 TAD (7000 /APT/
3645 3762# DCA APTN05 /APT/
3646 1373 TAD (7000 /APT/
3647 3761# DCA APTN06 /APT/
3650 1360 TAD (APTJ50 /APT/MODIFY SOME LOCS TO: JMP .+N.
3651 0357 AND (177 /APT/
3652 1356 TAD (5200 /APT/
3653 3755# DCA APTJ00 /APT/
3654 1354 TAD (APTJ51 /APT/
3655 0357 AND (177 /APT/
3656 1356 TAD (5200 /APT/
3657 3753# DCA APTJ01 /APT/
3660 1352 TAD (APTJ52 /APT/
3661 0357 AND (177 /APT/
3662 1356 TAD (5200 /APT/
3663 3751# DCA APTJ02 /APT/
3664 1350 TAD (APTJ53 /APT/
3665 0357 AND (177 /APT/
3666 1356 TAD (5200 /APT/
3667 3747# DCA APTJ03 /APT/
3670 1346 TAD (JMS I IAPTER /APT/MODIFY SOME LOCS TO: JMS I IAPTER.
3671 3745# DCA APT00 /APT/

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```

3672 1745# TAD APT00 /APT/
3673 3744# DCA APT01 /APT/
3674 1745# TAD APT00 /APT/
3675 3743# DCA APT02 /APT/
3676 1745# TAD APT00 /APT/
3677 3742# DCA APT03 /APT/
3700 1745# TAD APT00 /APT/
3701 3741# DCA APT04 /APT/

3702 1024 APTIZ0, TAD NOTTY /IS THERE A CONSOLE TERMINAL
3703 7650 SNA CLA /
3704 5740# JMP RUN0 /YES- START PROGRAM
3705 1737# TAD MINS0 /NO-SETUP TIME COUNTER
3706 3146 DCA MINS /SAVE THE TIME COUNTER
3707 1336 TAD (6003 /SETUP TO RUN ALL TESTS
3710 3076 DCA RUN08T /SAVE IT
3711 6002 IOF
3712 6224 RIF
3713 1335 TAD (6201
3714 3315 DCA .+1
3715 6201 CDF 0
3716 5734# JMP CHEX4+1 /GO RUN THE TEST

```

```

3734 0215
3735 6201
3736 6003
3737 1631
3740 1600
3741 1537
3742 1465
3743 1414
3744 1306
3745 1705
3746 4405
3747 1236
3750 1243
3751 1144
3752 1157
3753 0534
3754 0547
3755 0516
3756 5200
3757 0177
3760 0531
3761 0514
3762 0400
3763 1000
3764 0532
3765 0323
3766 0214
3767 0213
3770 4406
3771 4041
3772 7700

```

3773 7000
3774 4127
3775 1201
3776 0404
3777 1241
4000

```

PAGE                                /APT/

/APT/  ROUTINE TO 'NOTIFY' APT THAT THE PROGRAM IS RUNNING OK.

APTOK, 0                            /APT/
      IOF                          /APT/
      CLA                          /APT/
      TAD      APTIMX              /APT/DELAY 100MS.
      DCA      APTCTX              /APT/
      TAD      APTIMY              /APT/
      DCA      APTCTY              /APT/
      ISZ      APTCTY              /APT/
      JMP      =-1                 /APT/
      ISZ      APTCTX              /APT/
      JMP      =-5                 /APT/
      RIF                          /APT/AC=IF.
      TAD      (6201               /APT/CREATE A CDF INST.
      DCA      =+1                 /APT/MODIFY NEXT CDF INST.
      CDF      /APT/(MODIFIED CDF) DF=IF.
      CIF      70                  /APT/IF=FIELD 7.
      JMS      6500                /APT/CALL APT = 'PROG OK'.
      JMP I      APTOK             /APT/RTN FROM APT = RTN TO CALL+1.

4022 7771  APTIMX, -7              /APT/
4023 0000  APTIMY, 0                /APT/
4024 0000  APTCTX, 0                /APT/
4025 0000  APTCTY, 0                /APT/
  
```

/APT/ ROUTINE TO HANDLE ERRORS UNDER APT CONTROL.

```

4026 0000  APTER, 0                /APT/
4027 6002  IOF                      /APT/
4030 7200  CLA                      /APT/
4031 6224  RIF                      /APT/AC=IF.
4032 1377  TAD      (6201            /APT/CREATE A CDF INST.
4033 3236  DCA      =+3              /APT/MODIFY NEXT CDF INST.
4034 7240  CLA CMA                  /APT/
4035 1226  TAD      APTER            /APT/AC=ERROR PC.
4036 6201  CDF                      /APT/(MODIFIED CDF) DF=IF.
4037 6272  CIF      70              /APT/IF=FIELD 7.
4040 5775  JMP      6520            /APT/CALL APT = 'ERROR'.

/APT/  ROUTINE TO GET THE SPECIFICATIONS OF THE FIELDS TO BE TESTED
/APT/  (FIELD LIMITS). FIRST ATTEMPT IS FROM HCW1. IF THERE IS
/APT/  NO SPEC THERE THEN SPECS WILL BE TAKEN FROM THE SWITCH REGISTER
/APT/  (PSR OR HARD SR). IN EITHER CASE THE RESULT IS PRESERVED
/APT/  IN THE PSR. IF FIELD 7 IS SPECIFIED THEN FIELD 6 IS FORCED.
  
```

```

4041 0000  APTFL, 0                /APT/
4042 7200  CLA                      /APT/
4043 1021  TAD      HCW1             /APT/GET MEM SIZE FROM HCW1.
4044 0374  AND      (37              /APT/
4045 7012  RTR                      /APT/CONVERT TO HIGHEST FIELD LIMIT.
4046 0121  AND      K7              /APT/
4047 7450  SNA                      /APT/SKP IF VALID MEM SIZE WAS IN HCW1.
4050 5270  JMP      APTFL0           /APT/GO TRY SW REG FOR FIELD LIMITS.
4051 3224  DCA      APTCTX           /APT/FIELD 7 SPEC'D?
4052 1022  TAD      HCW2            /GET HARDWARE CONTROL WORD 2
4053 7700  SNA      CLA              /WAS APT SELECTED?
4054 5261  JMP      =+5              /NO-DO NOT CHECK FOR FIELD 7
4055 1224  TAD      APTCTX           /APT/
4056 1373  TAD      (7771            /APT/
4057 7650  SNA CLA                  /APT/SKP IF NO.
4060 7040  CMA                      /APT/FORCE FLD 6 AS HIGHEST FLD LIMIT.
4061 1224  TAD      APTCTX           /APT/
4062 3224  DCA      APTCTX           /APT/
4063 1020  TAD      PSR              /APT/
4064 0372  AND      (7700            /APT/
4065 1224  TAD      APTCTX           /APT/
4066 3020  DCA      PSR              /APT/PSR=FIELD LIMITS.
4067 5641  JMP I      APTFL         /APT/RTN TO CALL+1.

4070 7200  APTFL0, CLA              /APT/
4071 4440  GETSR                    /GET FIELD LIMITS FROM SWITCH REGISTER
4072 0371  AND      (77              /APT/
4073 3224  DCA      APTCTX           /APT/SEPARATE & TEMP STORE LO & HI LIMITS.
4074 1224  TAD      APTCTX           /APT/
4075 0370  AND      (7              /APT/
4076 3225  DCA      APTCTY           /APT/TEMP STORE HI LIMIT.
4077 1224  TAD      APTCTX           /APT/
4100 0367  AND      (70              /APT/
4101 3224  DCA      APTCTX           /APT/TEMP STORE LO LIMIT.
4102 1022  TAD      HCW2            /GET HARDWARE WORD 2
4103 7700  SNA      CLA              /WAS APT SELECTED?
4104 5321  JMP      NOTAPT           /NO-DO NOT CHECK FOR FIELD 7
4105 1224  TAD      APTCTX           /APT/FLD 7 SPEC'D AS LO LIMIT?
4106 1366  TAD      (7710            /APT/
4107 7640  SZA CLA                  /APT/SKP IF YES.
4110 5313  JMP      =+3              /APT/
4111 1365  TAD      (60              /APT/FORCE FLD 6 AS LO LIMIT.
4112 3224  DCA      APTCTX           /APT/
4113 1225  TAD      APTCTY           /APT/FLD 7 SPEC'D AS HI LIMIT?
4114 1373  TAD      (7771            /APT/
4115 7640  SZA CLA                  /APT/SKP IF YES.
4116 5321  JMP      =+3              /APT/
4117 1364  TAD      (6              /APT/FORCE FLD 6 AS HI LIMIT.
4120 3225  DCA      APTCTY           /APT/
4121 1020  NOTAPT, TAD      PSR       /APT/
4122 0372  AND      (7700            /APT/
4123 1224  TAD      APTCTX           /APT/
4124 1225  TAD      APTCTY           /APT/
4125 3020  DCA      PSR              /APT/PSR=FIELD LIMITS.
  
```

```

4126 5641      JMP I  APTFL      /APT/RTN TO CALL+1.
          /CB/  ROUTINE TO SAVE PAGE 37 OF FIELD 1
4127 0000      C88M,  0
4130 7200      CLA
4131 6224      RIF
4132 1377      TAD      (6201      /READ THE INSTRUCTION FIELD
4133 3342      DCA      C88H0      /ADD CDF 0 TO IT
4134 1363      TAD      (7577      /MODIFY THE CDF INSTR AT LOC C88M0
4135 3010      DCA      10      /SET UP PAGE 37 POINTER -1
4136 1362      TAD      (C88A-1    /SAVE IN AUTO INDEX 10
4137 3011      DCA      11      /GET ADDRESS -1 OF STORAGE AREA
4140 6211      C88M1,  CDF  10    /SAVE IN AUTO INDEX 11
4141 1410      TAD  I  10      /CHANGE DATA FIELD TO 1
4142 6201      C88M0,  CDF  10      /GET THE WORD
4143 3411      DCA  I  11      /CHANGE DATA FIELD TO PRG FIELD
4144 1010      TAD      10      /SAVE IN STORE AREA
4145 7040      CMA
4146 7640      SZA  CLA      /CHECK TO SEE IF PAGE DONE
4147 5340      JMP      C88M1      /DONE SAVING PAGE
4150 5727      JMP  I  C88M      /NO-DO NEXT WORD
          /YES-RETURN TO CALL+1

```

```

4151 4323      SRMSG, TEXT      "#SR#"
4152 2275
4153 0000
4154 7743      GESTMK, TEXT      "#?"
4155 0000
4156 3603      UPARRC, TEXT      "#C#"
4157 4300
4160 3607      UPARRG, TEXT      "#G#"
4161 4300

```

```

4162 4777
4163 7577
4164 0006
4165 0060
4166 7710
4167 0070
4170 0007
4171 0077
4172 7700
4173 7771
4174 0037
4175 6520
4176 6500
4177 6201
4200 4200      PAGE      /APT/

```

```

          /CB/  ROUTINE TO RESTORE PAGES 37 OF FIELD 0 AND 1
4200 7200      C88M,  CLA
4201 4426      PRNTMS      /GO PRINT UPARROW C
4202 4156      UPARRC      /POINTER TO MESSAGE

```

```

4203 6224      RIF
4204 1377      TAD      (6201      /GET THE PRESENT DATA FIELD
4205 3216      DCA      C88H0      /GET THE CDF INSTRUCTION
4206 1376      TAD      (7577      /SAVE THE NEW CDF INSTRUCTION
4207 3010      DCA      10      /SET UP AUTO INDEX FOR RESTORE OF 0
4210 1375      TAD      (C88A-1    /SAVE IN AUTO INDEX 10
4211 3011      DCA      11      /SETUP STORAGE POINTER
4212 1376      TAD      (7577      /SAVE IN AUTO INDEX 11
4213 3012      DCA      12      /SEUP AUTO INDEX OF RESTORE OF FIELD 1
4214 1376      TAD      (7577      /SAVE IN AUTO INDEX 12
4215 3013      DCA      13      /SETUP NEXT POINTER
4216 6201      C88M0,  CDF  10      /SAVE IN AUTO INDEX 13
4217 1010      TAD      10      /MODIFIED CDF INSTRUCTION TO PRG FIELD
4220 7040      CMA      /RESTORATION DONE
4221 7450      SNA
4222 5235      JMP      C88M1      /SKIP IF NO
4223 7621      7621      /DONE-GO TO MONITOR AT 7600
4224 1410      TAD  I  10      /CLEAR AC AND MQ
4225 7421      7421      /GET DATA FROM PROGRAM FIELD
4226 1411      TAD  I  11      /PUT IT IN THE MQ
4227 6211      CDF  10      /GET DATA TO BE PUT IN FIELD 1
4230 3413      DCA  I  13      /CHANGE DATA FIELD TO 1
4231 7521      7521      /PUT IT IN FIELD 1
4232 6201      CDF  00      /SWAP AC AND MQ
4233 3412      DCA  I  12      /CHANGE DATA FIELD TO 0
4234 5216      JMP      C88M0      /RESTORE FIELD 0 PAGE 37
4235 6203      C88M1,  CDF  CIP      /GO DO NEXT WORD
4236 5637      JMP  I  10+1      /CHANGE DATA AND INSTR FIELD TO 0
4237 7600      7600      /GO TO 7600 OF THAT FIELD
          /MONITOR STARTING ADDRESS

```

/ROUTINE TO GET VALUE OF SWITCH REGISTER USED

```

4240 0000      SRGET,  0
4241 7300      CLA  CLL
4242 1021      TAD      MCW1      /GET HARDWARE CONTROL WORD 1
4243 7710      SPA  CLA      /PSEUDO OR HARDWARE SWITCH REGISTER
4244 7614      7614      /USE HARDWARE
4245 1020      TAD      PSR      /USE PSEUDO SWITCH REGISTER
4246 5640      JMP  I  SRGET      /RETURN WITH SR VALUE IN AC

```

/ROUTINE USED FOR CONSOLE SWITCH REGISTER CHANGES

```

4247 0000      PSEUDO, 0
4250 7200      CLA
4251 4441      CHKCON
4252 5647      JMP  I  PSEUDO      /WAS CONSOLE ACTIVE
4253 4426      SRDEST, PRNTMS      /NO-RETURN BACK TO PROGRAM
4254 4151      SRMSG
4255 4440      GETSR      /PRINT SR QUESTION
4256 4434      PRNT4      /POINTER TO MESSAGE
4257 7346      CLA  CLL  CMA  RTL      /GET THE VALUE OF THE SWITCH REGISTER
4260 3333      DCA      TTYCNT      /PRINT THE 4 DIGITS
4261 1374      TAD      (CHARR0      /SETUP A COUNTER TO ACCEPT 4 DIGITS
4262 3265      DCA      CHGCHR      /SAVE THE COUNTER
4263 4425      L1SN      /GET POINTER FOR FIRST CHARACTER
          /SAVE THE POINTER FOR DIGITS
          /WAIT FOR KEYBOARD INPUT

```

```
4264 0001      1 /CHECK FOR A OCTAL DIGIT
4265 4303      /THIS LOCATION WILL GET MODIFIED
4266 7566      CHGCHR, CHARR0 =212 /CHECK FOR LINE FEED
4267 0200      START /LINE FEED TYPED= RETURN TO START
4270 7563      =215 /CHECK FOR CARRIAGE RETURN
4271 4317      RETYPE /RETYPE SR AND CONT IF DIGITS TYPED
4272 7575      =203 /CHECK FOR A CONTROL C
4273 4200      C8RM /CONTROL C TYPED =RETURN TO MONITOR
4274 7555      =223 /CHECK FOR A CONTROL S
4275 4364      CNTRS /WAS CONTROL S WAIT FOR "Q OR "C
4276 0000      0 /NONE OF ABOVE CHARACTERS=ILLEGAL CHAR
4277 4300      .+1 /GO TO NEXT ADDRESS TO PRINT ?
4300 4426      PRNTMS /GO PRINT ?
4301 4154      GESTMK /POINTER TO ? MESSAGE
4302 5253      JMP SRQEST /RETURN AND ASK QUESTION AGAIN
4303 3020      CHARR0, DCA PSR /SAVE THE LEAST SIGNIFICANT BIT
4304 1373      TAD [CHARR1 /UPDATA POINTER FOR CHARACTERS 2 3 4
4305 3265      DCA CHGCHR /SAVE THE POINTER ADDRESS
4306 5263      JMP CHGCHR-2 /RETURN FOR NEXT CHARACTER INPUT
4307 3332      CHARR1, DCA SAVCHR /SAVE THE CHARACTER TYPED
4310 1020      TAD PSR /GET THE VALUE OF SR
4311 7106      CLL RTL /MOVE IT INTO NEXT POSITION
4312 7004      RAL
4313 1332      TAD SAVCHR /ADD NEW CHARACTER TO IT
4314 3020      DCA PSR /SAVE THE NEW VALUE
4315 2333      ISZ TTYCNT /DONE ALL 4 CHARACTERS
4316 5263      JMP CHGCHR-2 /NO GET NEXT INPUT FROM KEYBOARD
4317 1374      RETYPE, TAD [CHARR0 /GET POINTER TO SEE IF SR ECHOED
4320 7041      CIA /NEGATE THE POINTER
4321 1265      TAD CHGCHR /GET THE POINTER STORED
4322 7650      SNA CLA /ECHO VALUE OF SR?
4323 5647      JMP I PSEUDO /NO-ONLY CR WAS TYPED=USE ORIGINAL VALUE
4324 4426      PRNTMS /RE-ECHO VALUE TYPED
4325 4151      BRMSG /POINTER TO SR MESSAGE
4326 4440      GETSR /GET VALUE OF SR
4327 4434      PRNT4 /PRINT THE 4 OCTAL DIGITS
4330 4437      CRLF /ISSUE A CR AND LF
4331 5647      JMP I PSEUDO /RETURN TO PROGRAM

4332 0000      SAVCHR, 0
4333 0000      TTYCNT, 0
```

/ROUTINE TO CHECK FOR CONSOLE RECEIVE FLAG

```
4334 0000      CHKKSF, 0
4335 6031      KSF /SKIP ON CONSOLE RECEIVE FLAG
4336 5734      JMP I CHKKSF /RECEIVE FLAG NOT SET RETURN TO PROGRAM
4337 4441      CHKCON /CHECK TO SEE IF CONSOLE WAS ACTIVE
4340 7410      SKP /NOT ACTIVE-CLEAR FLAG AND RETURN
4341 5344      JMP .+3 /ACTIVE CONSOLE-CHECK FOR "C OR "G
4342 6032      KCC /CLEAR CONSOLE RECEIVE FLAG
4343 5734      JMP I CHKKSF /RETURN TO PROGRAM
4344 4425      LISN /CHECK THE KEYBOARD CHARACTER
4345 7575      =203 /CODE FOR "C
```

```
4346 4200      C8RM /WAS A CONTROL C=EXIT TO MONITOR
4347 7571      =207 /CODE FOR "G
4350 4360      CNTRLG /WAS "G ECHO CHAR=ENTER SR QUESTION
4351 7555      =223 /CHECK FOR A CONTROL S
4352 4366      CNTRS1 /WAS A CONTROL S WAIT FOR "Q OR "C
4353 0000      0 /CHAR WAS NOT "C OR "G
4354 4355      .+1 /ECHO CHAR AND QUESTION MARK
4355 4426      PRNTMS /PRINT ? AND CR LF
4356 4154      GESTMK /POINTER TO MESSAGE
4357 5734      JMP I CHKKSF /RETURN TO PROGRAM

4360 4426      CNTRLG, PRNTMS /PRINT "G AND CR LF
4361 4160      UPARRG /POINTER TO MESSAGE
4362 4247      JMS PSEUDO /GO ASK THE SR QUESTION
4363 5734      JMP I CHKKSF /RETURN TO THE PROGRAM

4364 4772      CNTRS, JMS WAITOC /GO WAIT FOR A CONTROL Q OR C
4365 5263      JMP CHGCHR-2 /GO WAIT FOR NEXT CHAR

4366 4772      CNTRS1, JMS WAITOC /WAIT FOR A CONTROL Q OR C
4367 5734      JMP I CHKKSF /RETURN TO PROGRAM

4372 4710
4373 4307
4374 4303
4375 4777
4376 7577
4377 6201
4400          PAGE

4400 0000      FILLER, 0 /SET TO NUMBER OF FILLERS REQUIRED

/INPUT ONE OCTAL NUMBER TO AC 9 THRU 11
/GOOD RETURN IS JMS+2

4401 0000      ONEOCK, 0 /CALL BY "ONEOCK"
4402 4425      LISN
4403 0001      1
4404 4407      .+3
4405 0000      0
4406 4410      .+2
4407 2201      ISZ ONEOCK
4410 5601      JMP I ONEOCK

/INPUT TWO OCTAL NUMBERS TO AC 6 THRU 11
/GOOD RETURN IS JMS+2

4411 0000      TWOOCK, 0 /CALL BY "TWOOCK"
4412 4201      JMS ONEOCK
4413 5611      JMP I TWOOCK
4414 7104      CLL RAL
4415 7006      RTL
```

```

4416 3224      DCA   XPRNT2
4417 4201      JMS   ONEOCK
4420 5611      JMP I  TWOOCK
4421 1224      TAD   XPRNT2
4422 2211      ISZ   TWOOCK
4423 5611      JMP I  TWOOCK

```

/PRINT THE TWO OCTAL NUMBERS IN THE AC 6 THRU 11

```

4424 0000      XPRNT2, 0          /CALL BY "PRNT2"
4425 3211      DCA   TWOOCK
4426 1211      TAD   TWOOCK
4427 7012      RTR
4430 7010      RAR
4431 4432      PRNT1
4432 1211      TAD   TWOOCK
4433 4432      PRNT1
4434 5624      JMP I  XPRNT2

```

/TYPE THE ASCII CHARACTER IN THE AC

```

4435 0000      XTYPE, 0          /CALL BY "TYPE"
4436 3251      DCA   CHAR          /SAVE THE CHARACTER
4437 4441      CHKCON          /CHECK FOR A ACTIVE CONSOLE
4440 7610      SKP   CLA          /CONSOLE INACTIVE-TYPE THE CHARACTER
4441 4777      JMS   CNTRL3       /CONSOLE ACTIVE-CHECK FOR CONTROL S
4442 1251      TAD   CHAR          /GET THE CHARACTER SAVED AND PRINT
4443 6046      TLS
4444 7200      CLA
4445 6041      TSP
4446 5245      JMP   .-1
4447 6042      TCF
4450 5635      JMP I  XTYPE

```

4451 0000 CHAR, 0

/TYPE A CR AND LF WITH NUMBER OF FILLERS
/AS DETERMINED BY LOCATION "FILLER"

```

4452 0000      XCRLF, 0          /CALL BY "CRLF"
4453 7200      CLA
4454 1134      TAD   K215
4455 4436      TYPE
4456 1200      TAD   FILLER
4457 7040      CMA
4460 3266      DCA   XORS
4461 1133      TAD   K212
4462 4436      TYPE
4463 2266      ISZ   XORS
4464 5262      JMP   .-2
4465 5652      JMP I  XCRLF

```

4466 0000 XORS, 0

/PRINT 2 SPACES

```

4467 0000      SPACX2, 0        /CALL BY "SPACE2"
4470 4426      PRNTMS
4471 4473      .+2
4472 5667      JMP I  SPACX2
4473 4040      0040
4474 0010      0010      /USED BY LISN

```

/COMPARE INPUT TO LIST FOLLOWING CALL
/INPUT ONE CHARACTER IF AC=0
/USE LAST INPUT IF AC NON ZERO

```

4475 0000      XLISN, 0          /CALL BY "LISN"
4476 7640      SZA CLA
4477 5325      JMP   LISN1        /USE LAST INPUT SINCE AC NOT ZERO
4500 6031      KSF
4501 5300      JMP   .-1
4502 6036      KRB
4503 0357      AND   K177
4504 1360      TAD   K200
4505 3267      DCA   SPACX2
4506 1267      TAD   SPACX2
4507 1361      TAD   M212
4510 7450      SNA
4511 5315      JMP   .+4          /IS IT A LPT?
4512 1116      TAD   M3          /YES
4513 7640      SZA CLA
4514 5317      JMP   .+3          /IS IT A CR?
4515 4437      CRLF              /NO
4516 5325      JMP   LISN1
4517 1267      TAD   SPACX2
4520 1376      TAD   (=223
4521 7650      SNA CLA          /GET THE CHAR
4522 5325      JMP   LISN1       /CHECK FOR A CONTROL S
4523 1267      TAD   SPACX2      /WAS IT A CONTROL S
4524 4036      TYPE              /YES=DO NOT ECHO CHARACTER
4525 1675      LISN1, TAD I XLISN /PRINT THE CHARACTER
4526 2275      ISZ   XLISN       /GET COMPARE VALUE
4527 7450      SNA
4530 5336      JMP   LISN3       /EXIT?
4531 7500      SNA              /YES
4532 5346      JMP   LISNUM
4533 1267      TAD   SPACX2
4534 7640      SZA CLA          /LOOK FOR OCTAL NUMBER
4535 5343      JMP   LISN2       /COMPARE
4536 3266      LISN3, DCA   XORS /EQUAL?
4537 1675      TAD I  XLISN     /NO
4540 3275      DCA   XLISN
4541 1266      TAD   XORS
4542 5675      JMP I  XLISN
4543 7200      LISN2, CLA       /AC IS ZERO UNLESS OCTAL NUMBER
4544 2275      ISZ   XLISN

```

```

4545 5325 JMP LISN1
4546 7200 LISNUM, CLA /LOOK FOR OCTAL NUMBER
4547 1267 TAD SPACX2
4550 1354 TAD M270
4551 7500 SNA /IS IT LESS THAN 0?
4552 5343 JMP LISN2 /NO, SO NOT AN OCTAL NUMBER
4553 1122 TAD K10
4554 7510 M270, SPA /IS IT GREATER THAN ZERO?
4555 5343 JMP LISN2 /NO, SO NOT A NUMBER
4556 5336 JMP LISN3
4557 0177 K177, 0177
4560 0200 K200, 0200
4561 7566 M212, 7566

```

/ROUTINE TO CHECK TO SEE IF CONSOLE ACTIVE
 /RETURN CALL+1 IF CONSOLE INACTIVE
 /RETURN TO CALL+2 IF CONSOLE ACTIVE

```

4562 0000 CONCHK, 0
4563 1022 TAD HCM2 /GET HARDWARE WORD 2
4564 0375 AND (400 /MASK TO CONSOLE BIT
4565 7650 SNA CLA /HAS CONSOLE ACTIVE
4566 5762 JMP I CONCHK /NO RETURN TO CALL PLUS 1
4567 2362 ISZ CONCHK /CONSOLE ACTIVE BUMP RETURN
4570 5762 JMP I CONCHK /RETURN TO CALL PLUS 2

```

```

4575 0400
4576 7555
4577 4675
4600

```

PAGE

/PRINT PACKED ASCII TEXT TERMINATED BY
 /SIX-BIT 00

```

4600 0000 MESAGX, 0 /CALL BY "MESSAGE"
4601 7200 CLA
4602 1600 TAD I MESAGX
4603 3240 DCA FOROCK
4604 2200 ISZ MESAGX /SET UP RETURN
4605 1640 TAD I FOROCK
4606 7012 RTR
4607 7012 RTR
4610 7012 RTR
4611 4216 JMS MESAGF
4612 1640 TAD I FOROCK
4613 4216 JMS MESAGF
4614 2240 ISZ FOROCK
4615 5205 JMP -10
4616 0000 MESAGF, 0
4617 0131 AND K77
4620 7450 SNA /TERMINATOR (00)?

```

```

4621 5600 JMP I MESAGX /YES
4622 1235 TAD M43
4623 7450 SNA /CRLF?
4624 5233 JMP .+7 /YES
4625 1236 TAD K3
4626 7510 SPA /200 OR 300
4627 1237 TAD K100 /300
4630 1135 TAD K240 /200
4631 4436 TYPE
4632 5616 JMP I MESAGF
4633 4437 CRLF
4634 5616 JMP I MESAGF
4635 7735 M43, 7735
4636 0003 K3, 0003
4637 0100 K100, 0100

```

/INPUT 4 OCTAL NUMBERS TO AC
 /GOOD RETURN IS CALL+2

```

4640 0000 FOROCK, 0 /CALL BY "FOROCK"
4641 4430 TWOCT
4642 5640 JMP I FOROCK
4643 7106 CLL RTL
4644 7006 RTL
4645 7006 RTL
4646 3254 DCA XPRNT4
4647 4430 TWOCT
4650 5640 JMP I FOROCK
4651 1254 TAD XPRNT4
4652 2240 ISZ FOROCK
4653 5640 JMP I FOROCK

```

/PRINT FOUR OCTAL NUMBERS IN AC 11 THRU 8 FOLLOWED
 /BY TWO SPACES

```

4654 0000 XPRNT4, 0 /CALL BY "PRNT4"
4655 3240 DCA FOROCK
4656 1240 TAD FOROCK
4657 7012 RTR
4660 7012 RTR
4661 7012 RTR
4662 4433 PRNT2
4663 1240 TAD FOROCK
4664 4433 PRNT2
4665 4433 SPACE2
4666 5654 JMP I XPRNT4

```

/PRINT THE OCTAL NUMBER IN AC 9 THRU 11
 /CALL BY "PRNT1"

```

4667 0000 XPRNT1, 0
4670 0121 AND K7
4671 1274 TAD K260
4672 4436 TYPE
4673 5667 JMP I XPRNT1

```

```

4674 0260 K260, 260

```

/ROUTINE TO CHECK FOR A CONTROL S WHILE TYPING OUT MESSAGES
/TO EXIT ROUTINE IF A CONTROL S WAS TYPED-A CONTROL Q OR C MUST BE
/INPUTTED ON THE KEYBOARD

4675	0000	CNTRL8, 0		
4676	6031	KSF		
4677	5675	JMP I CNTRL8		/SKIP ON CONSOLE KEYBOARD FLAG
4700	6034	KRS		/RETURN TO TYPE ROUTINE-FLAG NOT SET
4701	0377	AND	(177	/READ THE CHARACTER STATICALLY
4702	1376	TAD	(=23	/MASK TO 7 BIT ASCII
4703	7640	SZA	CLA	/CHECK FOR A CONTROL S
4704	5675	JMP I CNTRL8		/WAS IT A CONTROL S
4705	6032	KCC		/NO-RETURN WITH KEYBOARD FLAG STILL SET
4706	4310	JMS		/CLEAR KEYBOARD FLAG FROM *S
4707	5675	JMP I CNTRL8		/WAIT FOR CONTROL Q OR C
				/RETURN TO PRINT MESSAGE BEING TYPED
4710	0000	WAITQC, 0		
4711	6031	KSF		/ROUTINE TO WAIT FOR CONTROL Q OR C
4712	5311	JMP		/WAIT FOR A CONTROL Q OR C TO EXIT
4713	6036	KRS	=-1	/
4714	0377	AND	(177	/READ THE CHARACTER TYPED
4715	1375	TAD	(=3	/MASK TO 7 BIT ASCII
4716	7450	SNA		/CHECK FOR A CONTROL C
4717	5774	JMP	CORH	/WAS IT A CONTROL C?
4720	1373	TAD	(=7	/YES-RESTORE MONITOR AND RETURN
4721	7450	SNA		/CHECK FOR A LINE FEED CHARACTER
4722	5772	JMP	START	/WAS IT A LINE FEED
4723	1373	TAD	(=7	/YES GO RESTART THE PROGRAM
4724	7640	SZA	CLA	/CHECK FOR A CONTROL Q TO
4725	5311	JMP	WAITQC+1	/WAS IT A CONTROL Q
4726	5710	JMP I	WAITQC	/NO-WAIT FOR APPROPRIATE KEY
				/RETURN TO WHENCE IT CAME
4772	0200			
4773	7771			
4774	4200			
4775	7775			
4776	7755			
4777	0177			
	5000	PAGE		
5000	0000	C8SA, 0		/THIS PAGE USED TO SAVE PG 37 OF FIELD 1

S

0000	11111110	00000000	11101111	11111111	11111111	11111111	11111111	11111111
0100	11111111	11111111	11111111	11111111	11111110	00000000	00000000	00000000
0200	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0300	11111111	11111111	11111111	11100000	00000000	00111111	11111111	11111111
0400	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0500	11111111	11111111	11111111	11111111	11111111	11000000	00001111	11111111
0600	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0700	11111111	11111111	11111111	11111111	11111111	11111111	11100000	00000011
1000	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
1100	11111111	11111111	11111111	11111111	11111111	11111111	11000001	11111111
1200	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
1300	11111111	11111111	11111111	11110000	00000000	00000000	00000000	11111111
1400	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
1500	11111111	11111111	11111111	11111111	11111111	11111111	11111110	00000111
1600	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
1700	11111111	11111111	11111111	11111111	11111111	11111110	00000000	00000011
2000	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
2100	11111111	11111111	11111111	11111111	11111111	11111111	11000000	00000111
2200	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
2300	11111111	11111111	11111111	11111111	11111111	10000000	00000000	00111111
2400	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
2500	11111111	11111111	11111111	11111110	00000000	00000000	00000000	01111111
2600	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
2700	11111111	11111111	11111111	11111111	11111111	11100000	00000000	00000111
3000	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
3100	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
3200	11111111	11111111	11111111	11111100	00000000	00000000	00000000	00000000
3300	00000000	00000000	00000000	00000000	00000000	00000000	00000000	00000000
3400	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
3500	11111111	11111111	11111111	11111111	11110000	00000000	00000000	00000001
3600	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
3700	11111111	11111110	00000000	00001111	11111111	11111111	11111111	11111111

```

4000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
4100 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111

4200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
4300 11111111 11111111 11111111 11111111 11111111 11111111 11111111 00111111

4400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
4500 11111111 11111111 11111111 11111111 11111111 11111111 11111111 10001111

4600 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
4700 11111111 11111111 11111110 00000000 00000000 00000000 00000000 00111111
    
```

```

5000 10000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
5100 00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
    
```

5200
5300

5400
5500

5600
5700

6000
6100

6200
6300

6400
6500

6600
6700

7000
7100

7200
7300

7400
7500

7600
7700

```

ABOVE 1715 C88M0 4142 CMEX05 1120 GETSR 4440
ADDER 2250 C88M1 4140 CMEX06 1134 GETSRX 0040
ADDER1 1316 C8F0 6203 CMEX07 1144 GTF 6004
ADDER2 1424 C8F1 6213 CMEX0A 1006 HCW1 0021
ADDER3 1476 C8F2 6223 CMGCHR 4265 HCW2 0022
ADDER4 1550 C8F3 6233 CMKCAC 0041 HEAD1 0073
ADDERR 3471 C8F4 6243 CMKCON 4441 HEAD12 2510
ADDRY1 1307 C8F5 6253 CMKKSF 4334 HIGHST 2731
ADDRY2 1415 C8F6 6263 CMKSW3 1647 IAPTR 0005
ADDRY3 1466 C8F7 6273 CIF0 6202 IAPTRK 0006
ADDRY4 1540 CDF0 6201 CIF1 6212 INSANE 0074
APT000 2052 CDF1 6211 CIF2 6222 K10 0122
APT001 2055 CDF2 6221 CIF3 6232 K100 4637
APTCTX 4024 CDF3 6231 CIF4 6242 K177 4557
APTCTY 4025 CDF4 6241 CIF5 6252 K20 0123
APTE00 1705 CDF5 6251 CIF6 6262 K200 4560
APTE01 1306 CDF6 6261 CIF7 6272 K207 0132
APTE02 1414 CDF7 6271 CINT 6204 K212 0133
APTE03 1465 CHAR 4451 CNTRLG 4360 K215 0134
APTE04 1537 CHARR0 4303 CNTRL3 4675 K240 0135
APTER 4026 CHARR1 4307 CNTRS 4364 K245 2123
PTFL 4041 CMECK 3015 CNTRS1 4366 K260 4674
PTFL0 4070 CMECK0 3016 CNV 2024 K3 4636
PTIMX 4022 CMEXA 0214 CODERR 2201 K30 0124
PTIMY 4023 CMEXB 0253 CONCHK 4562 K340 2124
PTIZ 3600 CMEXC 0262 CONHLT 2400 K40 0125
PTIZ0 3702 CMEXC1 0274 COUNT 0112 K4060 0136
PTIZ1 3614 CMEXD 0302 CRLF 4437 K50 0126
PTJ00 0516 CMEXD1 0310 CRLFP 0037 K60 0127
PTJ01 0534 CMEXE 0317 CSS 3000 K6003 1636
PTJ02 1144 CMEXE2 0400 C88B 3010 K6060 2126
PTJ03 1236 CMEXM 0476 CUF 6264 K6201 0137
PTJ50 0531 CMEXM1 0503 DOWN 1745 K6203 1637
PTJ51 0547 CMEXM2 0510 ERR1 2272 K7 0121
PTJ52 1157 CMEXM3 0515 ERR1A 2307 K70 0130
PTJ53 1243 CMEXN 0532 ERR2 2323 K707 2125
APTN00 0213 CMEXN0 0533 ERR3 2441 K77 0131
APTN01 0214 CMEXN1 0600 ERR4 2437 KABOVE 0071
APTN02 0233 CMEXN2 0622 ENRL0C 0113 KBELOW 0072
APTN03 0532 CMEXN3 0640 ERRM 2475 KBINT 0100
APTN04 1000 CMEXN4 0656 ERRMES 2407 KDOWN 0110
APTN05 0400 CMEXN5 0674 ERROR0 2242 KHIGH 2752
APTN06 0514 CMEXN6 0712 ERROR1 2243 KSFCHK 0004
APTOK 4000 CMEXN7 0730 ESL 0102 L40 4448
APTOK0 1201 CMEXN8 0746 EXTAD0 0205 LAST 3234
BDATA 0105 CMEXN9 0761 FILLER 4400 LEGAL 0410
BELOW 1734 CMEX0 1000 FIRS 3233 LEGAL0 0075
C8RM 4200 CMEX00 1024 FIVE 1262 LEGAL1 0463
C8RM0 4216 CMEX01 1040 FOROCK 4640 LIMIT 2263
C8RM1 4235 CMEX02 1054 FOROCP 0031 L10N 4429
C8SA 5000 CMEX03 1070 FOROCT 4431 LISN1 4525
C8SM 4127 CMEX04 1104 GDATA 0106 LISN2 4543
    
```

/KM8E EXTENDED MEMORY ADDRESS TEST				MAINDEC-08-DHKMC-B-L	PAL10	V142A	9-APR-76	13150	PAGE 1-54	SEQ 0072
L18N3	4536	RDF	6214	STK7	0067		XLISP	0025		
L18N4	4546	RELERR	3525	STKPIN	0103		XMESAG	0140		
LOOP1	3200	RELO	1655	STKTST	0104		XORS	4466		
LOOP2	3207	RELO2	1674	STOP	2251		XPRN1P	0032		
LOOP2A	3214	RELO3	1676	SUP	6274		XPRN2P	0033		
LOOP2B	3216	RELO4	1701	SW0	0042		XPRN4P	0034		
M1	0114	RELOS	1712	SW1	0043		XPRNT1	4667		
M2	0115	RETK	3034	SW2	0044		XPRNT2	4674		
M212	4561	RETURN	2200	SW3	0045		XPRNT4	4654		
M270	4554	RETYPE	4317	SW4	0046		XRETUR	0143		
M3	0116	RIB	6234	SW5	0047		XSTXTY	0141		
M34	0120	RIF	6224	SW60	0130		XSTDP	0144		
M4	0117	RMF	6244	SW911	0121		XTYPE	4435		
M43	4635	RTF	6005	SWAD	3206		Z10	3462		
MESAGE	2042	RUN0	1600	SWAD0	3235		Z11	3463		
MESAGF	4616	RUN1	1605	YDF1	1275		Z20	3422		
MESAGP	0026	RUN2	1612	YDF2	1402		Z21	3423		
MESAGX	4600	RUN3	1617	YDF3	1445		Z22	3426		
MINS	0146	RUN4	1624	YDF4	1516		Z23	3431		
MINS0	1631	RUNTST	0076	TEMP	0111		Z24	3434		
MINS1	1632	SAME	1640	TEST	1200		Z0	2702		
MINS2	1633	SAVCHR	4332	TEST1	1273					
MINS3	1634	SETSW	2624	TEST1A	1274					
MINS4	1635	SETSW1	2644	TEST1B	1302					
MOVE	0107	SINT	6254	TEST2	1400					
MSRGHT	2067	SIXTY	2000	TEST2A	1403					
MS8L	1751	SIXTY0	2037	TEST2B	1410					
MTP	2121	SIXTY1	2040	TEST3	1443					
NOBELL	2215	SIXTY2	2041	TEST3A	1446					
NORELO	0070	SPACE2	4435	TEST3B	1456					
NO8TK	2152	SPACX2	4467	TEST4	1514					
NOTAPT	4121	SPCX2P	0035	TEST4A	1517					
NOTTY	0024	SRGET	4240	TEST4B	1530					
ONEOCK	4401	SRME6G	4151	TESTAD	0077					
ONEOCP	0027	SRQEST	4253	TITLE	2600					
ONEOCT	4427	SSL	0101	TOSL	3077					
PASCNT	1261	STACK0	0050	TST3Y3	3017					
PASME3	1263	STACK1	0051	TYVCNT	4333					
PERR1	3400	STACK2	0052	THOOCK	4411					
PERRM	3436	STACK3	0053	THOOCKP	0030					
PINF	2127	STACK4	0054	THOOCKY	4430					
PNOREL	2647	STACK5	0055	TYPE	4436					
PREL	2706	STACK6	0056	TYPECH	2070					
PREL1	2726	STACK7	0057	TYPEP	0036					
PRNT1	4432	START	0200	TYPESP	3047					
PRNT2	4433	STK0	0060	UPARRC	4156					
PRNT4	4434	STK1	0061	UPARRG	4160					
PRNTM3	4426	STK2	0062	WAITQC	4710					
PSEUDO	4247	STK3	0063	XADDER	0145					
PSR	0020	STK4	0064	XCODER	0142					
QUESTM	4154	STK5	0065	XCRLF	4452					
RELL	2206	STK6	0066	XLISN	4475					

ERRORS DETECTED: 0
LINKS GENERATED: 185
RUN-TIME: 13 SECONDS
3K CORE USED